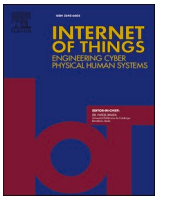
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Review article



Enhancing supply chain resilience and efficiency through internet of things integration: Challenges and opportunities

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ABSTRACT

This study explores the current challenges and future directions of Internet of Things (IoT) in supply chains, focusing on the drivers and barriers to its adoption. It starts with a review of 913 documents from the Web of Science, spanning 2009 – 2023, and narrows down to 408 relevant publications. Employing bibliometric analysis—descriptive, trend topic, conceptual, and network analyses—the research addresses the challenges at the intersection of IoT and supply chain. Findings highlight a surge in research from 2018, with worldwide contributions. The study identifies the 20 most active countries, top 10 journals, and key papers in the field. The result reveals a transition from studies predominantly focused on enhancing efficiency in the supply chain using IoT to an increased emphasis on resilience, particularly due to global disruptions like COVID-19, while also considering sustainability and digital transformation. Integrating machine learning and IoT to predict future conditions in supply chains marks also a novel approach. Based on the network analysis several critical challenges, including issues of security, privacy, interoperability, standardization, scalability, and energy efficiency, are identified as obstacles to effective IoT integration. The research also points to blockchain technology as a promising solution for addressing these challenges, facilitating decentralized trust, and enhancing cybersecurity. Moreover, the study emphasizes IoT's capacity for real-time tracking of logistics, production, and agri-food supply chains. A SWOT analysis outlines critical factors for integrating IoT in supply chain management, providing policy recommendations for industry practitioners and a framework for further investigation into IoT's potential and challenges in supply chains.

1. Introduction

Increased global uncertainty has led to more frequent disruptions in supply chains, making them vulnerable to various challenges. The COVID-19 pandemic, exemplified by social distancing measures, has significantly shaken up supply chain operations, resulting in shortages of raw materials and disruptions in production and transportation [1,2]. Diverse yield rates also introduce uncertainty in the production phase of the manufacturing supply chain, impacting overall efficiency and predictability [3]. Beyond these immediate challenges, forecasting, and managing issues like unsafe products, authenticity fraud, and product health has become essential in contemporary supply chain management [4]. Timely delivery of materials and products within specified periods and under predefined conditions is also crucial for effective supply chain operations [5]. The survival of supply networks hinges on careful anticipation and the formulation of workable business plans.

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A well-equipped supply chain, characterized by qualities such as resilience, speed, traceability, and reliability, is essential for effectively managing today's challenges [6]. Innovative technologies, including blockchain, cloud computing, and the Internet of Things (IoT), have been identified as pivotal in navigating the challenges facing modern supply chains [7]. Among these, the IoT stands out as a transformative paradigm, enhancing supply chain management through the integration of smart connected products. This technology is defined by its capacity to harness advancements in processing power, device miniaturization, and widespread wireless connectivity [8]. IoT establishes a seamless network of RFID (Radio-Frequency Identification) tags, sensors, actuators, and mobile phones. This network not only fosters collaboration but also propels supply chains towards achieving shared objectives more efficiently [9].

The IoT, operating at the supply chain level, enhances various aspects of supply chain operations, including processes, products, and physical assets like transportation and equipment [10,11]. Continuous data generation and real-time monitoring of physical objects contribute to controlled and easily accessible data, fostering cooperative decision-making and smooth integration [12,13]. The IoT's provision of extensive, current, and comprehensive data empowers supply chains to adjust to changing business conditions and make informed decisions [14,15]. Integrating IoT into the supply chain landscape has the potential to address longstanding industry challenges, including enhancing efficiency, environmental sustainability, and resilience [16,17].

However, the trajectory of IoT in supply chain management encounters restrictions and barriers. Challenges include the acquisition of technical infrastructure and skill sets, concerns about social and economic influences such as employment apprehensions and upfront costs, and the lack of a globally interoperable platform [9,18,19] Additionally, privacy and data security concerns, compounded by the decentralized nature of supply chain participants, pose significant challenges [20,21]. The data structure, compounded with privacy concerns, makes accessing, managing, and utilizing it a sophisticated process [22]. Knowledge asymmetry and potential information leakage also present additional challenges [21,23].

To navigate these challenges and illuminate the way forward, a comprehensive understanding of the challenges in the intersection of IoT and supply chain is essential. While existing studies have investigated various facets of IoT and Industry 4.0 within supply chains, there remains a noticeable gap in synthesizing these findings to clearly defines the complex challenges and opportunities present. This study aims to bridge this significant research gap. The primary goal of this research is to "outline the current challenges and future development trajectories of IoT in supply chains, with a specific focus on understanding the drivers and barriers affecting its adoption." Achieving this goal requires establishing a clear framework to analyze the intricate dynamics within the field. In this regard, we have formulated the following research questions:

RQ1. What is the current publication trend in IoT – supply chain and related challenges?
RQ2. Which are the most influential articles on IoT – supply chain and related challenges?
RQ3. What are the countries with the highest contribution to do ing research in the research field?
RQ4. What factors contributed to the evolution of research focused on the intersection of IoT and supply chain challenges, and what prospective trajectories are anticipated for its ongoing development?
RQ5. What factors serve as drivers and barriers in the integration of IoT within supply chain operations?

The structure of this paper is organized as follows: Section 2 establishes the foundation by outlining the materials and methods used in our study, including detailed approaches to data collection and analysis. Section 3 delves into our comprehensive bibliometric analysis, where the results of descriptive analysis address RQ1, RQ2, and RQ3. This section also presents how the findings from trend topic and conceptual analysis inform RQ4, while the outcomes of network analysis, including keyword co-occurrence and co-citation, address RQ5. Section 4 discusses these results, proposing a SWOT analysis to examine critical factors influencing the IoT-enabled supply chain ecosystem. Section 5 concludes the paper, summarizing key insights, acknowledging research limitations, and offering directions for future studies.

2. Material and method

In this section, we first discuss the data collection methodology and then elaborate on the methodology applied in the research.

2.1. Data collection

The search was conducted in the core collections of Web of Science using specific criteria. We focused on keywords related to the convergence of supply chain paradigms with Internet of Things (IoT) technology and its associated challenges. Our search query consisted of the following string: ("Internet of Thing*" OR "IoT*") AND "challenge*" AND "supply chain*". We sought papers that included these terms in their topic, title, or abstract. Including an asterisk (*) after each word enabled the consideration of spelling variations, such as "supply chains", "supply chain's", and "supply chains", to be considered in the search results.

Our most recent search, conducted on May 12, 2024, was limited to the English language papers published between January 1, 2009, and December 31, 2023. This search yielded 913 documents encompassing a variety of publication types, including articles, review articles, proceedings papers, early access materials, book chapters, and other miscellaneous items [24]. To refine our analysis, we restricted our selection utilizing the Web of Science categorizations as a qualitative measure. This process initially identified about 91fields across categories, which were streamlined into broader mega-categories for relevance and clarity. These encompassed engineering (e.g., electrical, electronic, industrial, manufacturing, environmental, multidisciplinary engineering), computer science (including information systems, theory and methods, interdisciplinary applications, artificial intelligence, software engineering,

hardware architecture, cybernetics), management and business (including operations research, management science, business, finance), environmental sciences and green technology, and agriculture and food sciences (including agricultural economics, policy, engineering, multidisciplinary agriculture, food science technology, and agronomy). This categorization allowed us to focus on 35 fields most pertinent to our research, enhancing the relevance and coherence of our analysis. Applying these filters reduced the number of papers to 854. We meticulously screened the selection, examining each paper's abstract to ensure it specifically addressed IoT in the context of the supply chain. This step involved a manual screening of the abstracts and resulted in the retention of 627 papers. The primary research objective of this study is to outline the current challenges and future development trajectories of IoT in supply chains. To ensure that all selected papers are eligible to contribute to this overarching goal, we applied an additional filter. We used Atlas.TI software renowned for its rich textual analysis capabilities and sophisticated search functionalities to review the abstracts for keywords like "challenge", "barrier", "issue", and "obstacle", along with about 13 additional synonyms including "risk", "dilemma", "predicament", "impediment", "trouble", "battlement", "deterrent", "impediment", "resistance", "obstruction", "quandary", and "block". This refined the selection to a final count of 408 papers. Fig.1 illustrates the resource selection process.

Subsequently, we downloaded the data in TXT format for further analysis using Biblioshiny, the web app of the R-package Bibliometrix [25]. This tool can produce clear, insightful, and significant insights in the context of research [26]. We also used VOSviewer, a freely accessible Java-based software, to analyze data and construct large bibliometric networks based on natural language processing and text mining algorithms.

2.2. Data analysis

This section provides detailed information on the data analysis methodologies and the tools used to conduct this analysis.

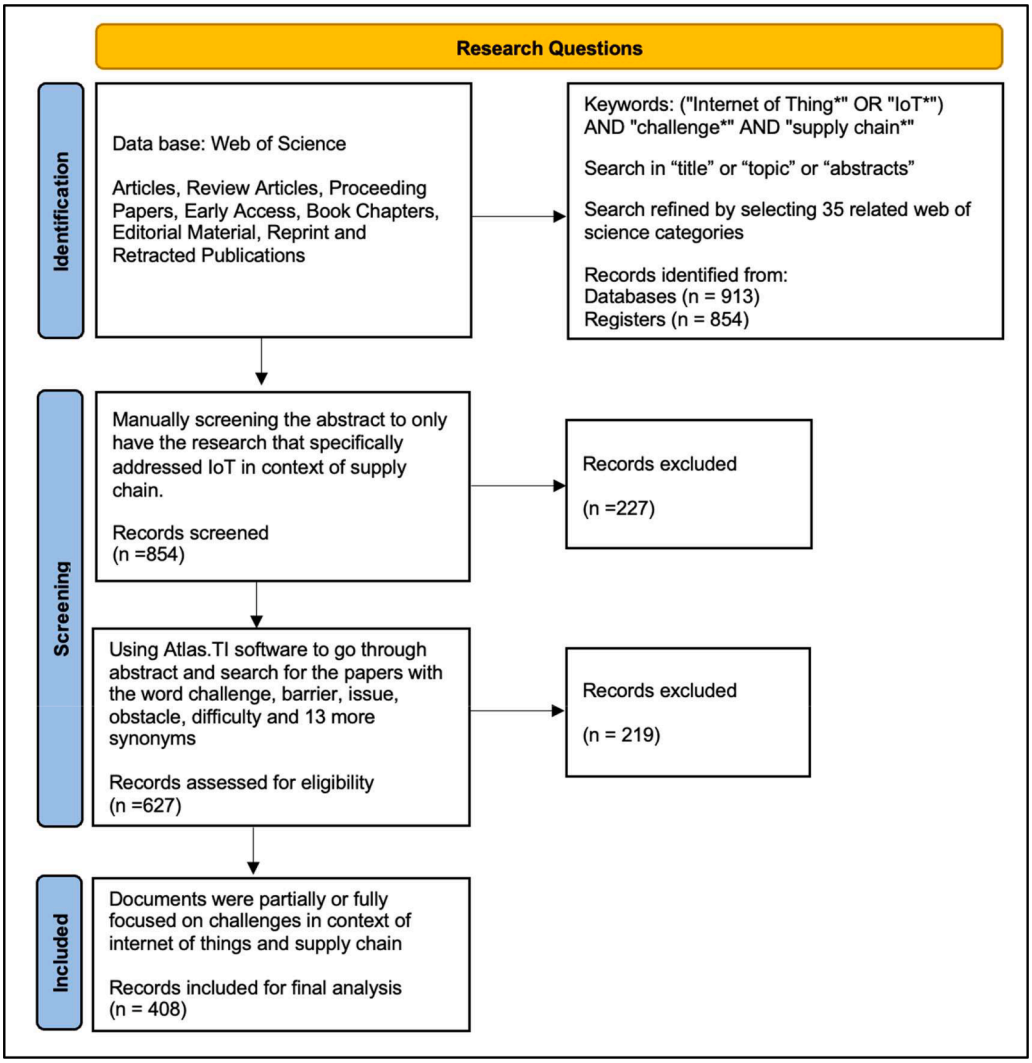


Fig. 1. Prisma flowchart.

Bibliometric analysis is employed as the primary analytical framework, incorporating descriptive, trend topic, conceptual, and network analysis to address the research questions. Bibliometric analysis is a structured method used to quantitatively evaluate the vast body of academic literature, offering insights into publication patterns, research trends, and the impact of scholarly work [27]. This study utilizes well-known practices to conduct an in-depth and reliable systematic analysis of challenges in the intersection of IoT and supply. A variety of data visualization techniques are utilized to present the data clearly and concisely. For descriptive, trend topic and conceptual analyses, we employed the R-package Biblioshiny and for network analysis we used the visualization software VOSviewer.

2.2.1. Descriptive analysis

The implementation of descriptive analysis enabled us to concisely encapsulate and present the fundamental characteristics inherent in the dataset. We use temporal distribution of the research, most relevant resources, most local cited resources, most local and most global citation, most local cited references and the status and trend of counties publication as the indicators for answering the first three questions in the research. Document and source investigation of Biblioshiny, serves as the principal instrument for our analysis in this part.

2.2.2. Trend topic analysis

We undertook the combination of trend topic analysis and a conceptual analysis of thematic evolution, for addressing the fourth research question. Our aim was to unravel the underlying thematic within the dataset in different period. To achieve this, we employed the features available in Biblioshiny.

Trend topic analysis is a methodological approach designed to chart the evolution of research themes over time, identifying how frequently specific topics appear within a body of literature. This technique involves assigning a representative year to each topic, typically determined by the median of the distribution of the topic's occurrences. The primary goal is to create a graphical representation that illustrates the temporal distribution of topics, facilitating an understanding of how research interests shift and emerge. The trend topic map shows a scatter plot where the horizontal axis represents time, and the vertical axis lists the most frequently occurring terms. The gray bar indicates the first and third quartiles of the occurrence distribution. Each point or "bubble" on this plot corresponds to the prevalence of a topic in its median year, with the bubble's size proportional to the term's occurrence frequency [28].

Implementing this analysis commenced with the extraction of keywords from the literature via the Biblioshiny. After this extraction, a process of keyword refinement including the harmonization of terminology and acronyms (replacing "Internet of Things" by its abbreviated form "IoT") was conducted to guarantee uniformity among the terms.

2.2.3. Conceptual analysis

The conceptual analysis is useful in forming clusters and creates a comprehensive overview of varying research directions within a specific academic domain [29]. This analysis is based on this idea that occurrence of certain keywords in documents that are sharing them means that these documents are similar and originate from a specific research niche [30,31]. Keywords appearing together indicate a network of co-occurring terms, suggesting thematic similarities. Walktrap clustering algorithm assists highlighting the distinct subject area within a domain on this network. The keywords are assessed through an examination of Callon centrality and density, where centrality indicates a theme's significance across the research landscape, and density reflects how developed the theme is [30,32–35]. Each cluster within the matrix represents a theme, with the most frequently occurring word within the cluster naming the bubble, and the bubble's size reflecting the frequency of occurrences within the cluster [28]. Clusters are classified as four themes including basic themes, motor themes, emerging or declining themes, and niche themes. For conducting the conceptual analysis in this study, the Biblioshiny [25] was used to analyze the author's keywords thematic progression, considering 250 keywords, setting the minimum cluster frequency at 5, and weighting the inclusion index by word occurrences at 0.1 for clear and interpretable outcomes, plotting one keyword per cluster. By dividing the study period into segments, it was possible to trace and depict the evolution of topics over time. A single division made in 2022 yielded two thematic maps: one depicting 2009–2021 and the other one for 2022–2023.

2.2.4. Network analysis

In next step of our analysis, we use network analysis and explore the use of keywords co-occurrence and co-citation networks to answer the fifth research question. These techniques play a crucial role in mapping out the connections among different research topics. They help identify key themes and provide a deeper understanding of the field's framework. This understanding aids in recognizing groups of related subjects and highlighting those that are most influential.

Examining keyword co-occurrence networks shows how to link study subjects [30,36]. The VOSviewer is used to analyze keyword co-occurrence. The VOSviewer offers the chance to obtain results based on the co-occurrence of keywords, keyword plus, and author's keywords. In this analysis, we use the author's keywords. The minimum number of occurrences of the keyword is set equal to 3. In first analysis 111 keywords out of 1183 met the threshold. A filter was performed to replace the synonyms and eliminate the unrelated keywords. As a result, 95 out of 1152 keywords met the threshold.

To evaluate the confluence of challenges arising from the intersection of IoT and supply chain realms, we conducted an insightful co-citation analysis using VOSviewer. Co-citation entails the instances where two documents are jointly cited within other scholarly works [37]. Through this methodology, we harnessed network visualization techniques to elucidate the interrelationships among these papers in terms of their co-citation patterns. Within this visualization, the proportions of labels and circles associated with each item are contingent upon the corresponding item's citation count. Notably, items with higher citation counts are represented by more sizable labels and circles. Our unit of analysis comprised the cited references, with a stipulated minimum occurrence threshold of 10

citations per reference. This selective criterion culled a focused set of approximately 119 cited references from an initial pool of 23,806. In the visual representation, the connections between items are delineated by lines, while the spatial proximity of nodes in the visualization indicates the degree of co-citation linkage between papers. The juxtaposition of two papers in proximity signifies a stronger co-citation-related affiliation.

Based on co-citation result, an approach is employed to conduct an analysis across all clusters of papers. Within the context of this inductive analysis, papers with high citations are selected. The entire paper corpus is meticulously examined through a scanning process that unfolds in distinct steps. During the paper scanning process, the primary contributions of each paper are thoroughly investigated as a starting point. Following this, an assessment is made regarding how the paper's content revolves around the realms of IoT and supply chain. Finally, an exploration into the types of challenges discussed within the paper is undertaken. Drawing from the outcomes of this analysis, the principal themes and concerns unique to each cluster are extracted.

Through the application of this multifaceted methodology, our objective is to attain an exhaustive comprehension of the dataset. By peering into its intricacies from various angles, we strive to contribute a reservoir of invaluable insights to the broader research landscape.

3. Results

In this section, we present the results of the bibliometric analysis. For each part, we first display the findings and then connect them to the research questions.

3.1. Descriptive result

Below is the result of descriptive analysis. This analysis allows us to assess the publication status, the influential publications, and the most active countries in the research context.

3.1.1. Assessment of publication status

In the first step, we assessed the temporal distribution of the articles, the most relevant sources, the number of publications they cover, and the most local cited sources. Fig. 2 illustrates the temporal distribution of research in the field of IoT challenges in the context of the supply chain. The number of global publications per year is illustrated in this figure. It demonstrates a noticeable upward trend, indicating an increasing number of studies over time. While the research count remained stable between 2009 and 2018, there has been a significant surge since 2018, with the number of articles growing from approximately 14 papers in 2018 to 117 in 2022. This uptrend in research at the junction of IoT and supply chain, along with its associated challenges, underscores the field's growing significance and highlights the value of conducting research within this domain.

Fig. 3 shows the most relevant sources (journals, books, others) that published papers related to the challenges in the context of IoT and supply chain. The presence of the "Sustainability" journal in the first place of relevance of sources in Fig. 3, is an evident of the significance of this topic in research context [16,38,39]. This demonstrates a rising interest in incorporating IoT technologies to support sustainable supply chain practices. Additionally, the featuring of "Sensors" as a journal that focuses on the integration of sensor technologies and data collection, underscores the pivotal importance of sensor technologies [40,41]. The challenges for developing and applying the sensors may be one of the most critical challenges in applying IoT in supply chain. Furthermore, as Fig. 3 shows journals such as "IEEE Access" [42] "Computer and Industrial Engineering" [43,44] and "Applied Sciences-Basel" [45] are ranked as the top relevant sources which show a comprehensive investigation of IoT applications in streamlining supply chain operations.

Fig. 4 shows the most locally cited sources graphs. This graph shows the number of times a source is cited in the selected data set.

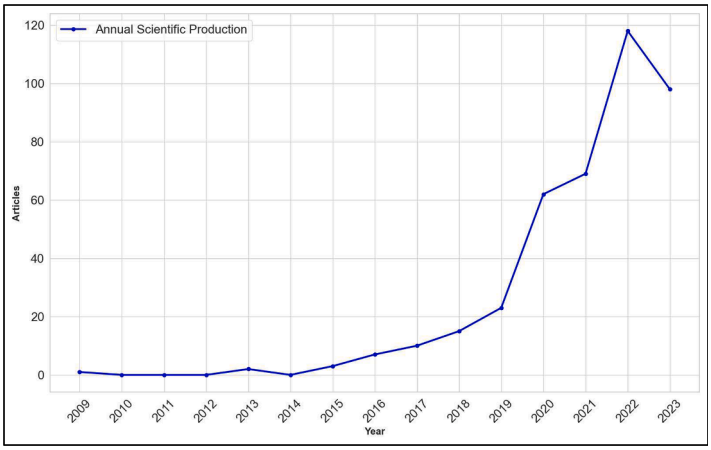


Fig. 2. Temporal distribution of research.

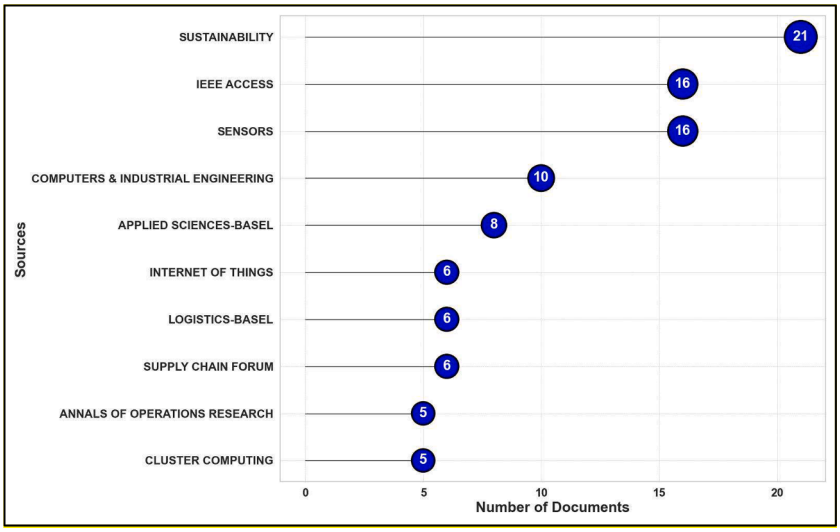


Fig. 3. Most relevant source.

"IEEE ACCESS" [46–48] spans a comprehensive array of topics in computer science and electrical engineering, making it a prime venue for detailed explorations into the intersections of supply chain management and IoT, including but not limited to, efficiency optimization, real-time tracking, data analytics, and cybersecurity within this framework. Research focusing on the integration challenges of IoT within supply chain management has frequently cited journals such as "Journal of Cleaner Production", "The International Journal of Production Research", and "International Journal of Production Economics". As a result of this focus, these publications are considered as pivotal sources, contributing significantly to the discourse on the complexities of integrating IoT technologies with production supply chain processes. Based on sources analysis, it can be concluded that researchers approach the study of IoT, supply chain, and related challenges through diverse lenses. Some delve into the intricacies of production processes in supply chain, while others focus on how it can influence sustainability and economic dimensions of the supply chain. Specific attention is given to sensor technologies for optimizing supply chains. Additionally, specialized studies explore the application of IoT in production and agricultural supply chain operation and providing unique insights into the technology integration and impacts in these industries.

3.1.2. Review of the influential publications

The assessment of documents and their citations is conducted based on three key aspects including global citation, local citation, and most locally cited references.

The "global citation" ranking, evaluates the total number of citations a paper has received across the entire academic landscape, providing an overview of its broader impact. Fig. 5 illustrates the documents with the most global citations. In terms of global citation

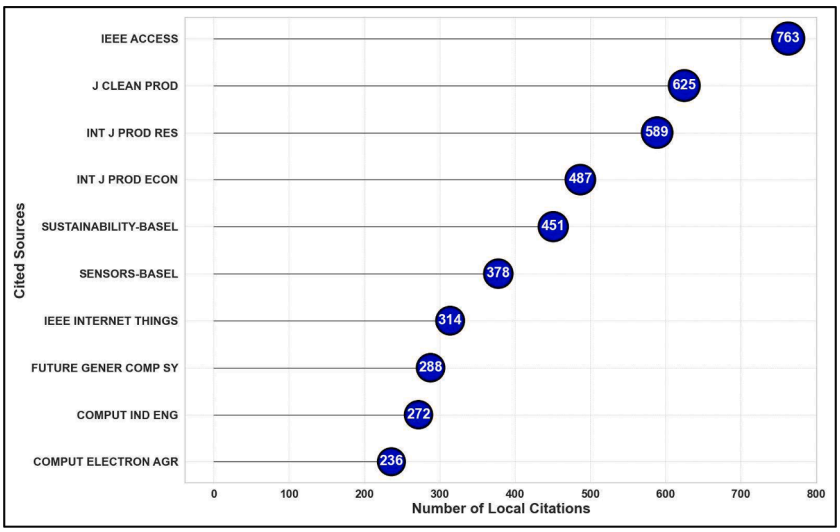


Fig. 4. Most local cited sources.

rank, Wolfert et al., 2017 [49] has the first place. This paper discusses the digital transformation in agriculture through data, technology, and communication integration and indicates its pivotal role in smart farming and supply chain advancements. Following this, Q. Feng et al., 2019 's work [50] on privacy protection in blockchain systems is highlighted, showing its importance in secure communication and data management beyond IoT.

The "local citation" ranking is equally important as it ranks papers based on the number of citations they receive from other papers within the examined dataset. This localized approach is significant because it highlights the paper's direct relevance and influence within a specific field of study, offering a more focused perspective on its impact. It helps identify which research works are most valued by the community strongly associated with the topic of interest, thus complementing the global citation ranking by showing a detailed picture of the paper's influence in its niche themes. Fig. 6 illustrates the documents with the most local citations. In assessment of the local citation, H. Feng et al., 2020a [51] stands out for having the highest ranking in local citation documents. This research explores blockchain's application in agri-food traceability, emphasizing its significance in the agricultural domain.

The "most locally cited references" segment emphasizes the most influential references within the examined dataset, further underlining the importance of localized impacts in scholarly communications [49]. Fig.7 illustrates the result of this analysis. As this figure highlights, the "Internet of Things and Supply Chain Management: A Literature Review" by Ben-Daya et al., 2022a [6], is the most local cited reference. This study reviews the constructive collaboration between IoT and blockchain in creating competitive supply chain advantages. Together, these studies chart a clear path from broad digital agricultural innovations towards specialized technological solutions, pinpointing blockchain integration, privacy, and supply chain efficiency as pivotal areas in current research.

3.1.3. Active countries in IoT-supply chain challenges context

The scientific output of nations within the context of IoT-supply chain challenges is depicted in Fig. 8. The blue spectrum in this figure indicates the number of publications related to each country. Darker blue represents a higher number of publications, while lighter blue indicates fewer publications. Additionally, the names of the top 20 countries in terms of publication count and their respective publication numbers are specified in this figure. India and China with 150 and 136 scientific productions are the notable contributors, respectively, demonstrating their active engagement in examining the challenges at the junction of IoT and supply chains. The USA and the UK show their active participation with 99 and 71 productions. Australia is the fifth country it terms of publication's count with 49 citations.

Fig.9 presents an analysis of research trends in IoT-enabled supply chains across the five countries with the highest number of publications. India and China stand out as a stunning example of research activity with an exponential growth. Starting with a low presence, both countries increased their involvement, in recent years. This shows a significant strategic shift in research priorities, possibly motivated by awareness of the potential transformational impact of IoT on supply chains. The United States, the United Kingdom and Australia showed a steady increase in research participation, indicating a smooth and ongoing commitment to investigating IoT-supply chain intersections. The findings highlight that nations' research tendencies are changing rapidly. Both quick rise and gradual development demonstrate the effect of IoT in revolutionizing the supply chain issues on a global scale. The diverse approaches undertaken by these countries reveal a range of policies in action to address the evolving challenges of IoT-driven supply chain advancements.

In summary, the descriptive analysis section provides a concise overview of publication trends (RQ1), key papers (RQ2), and the countries contributing most significantly (RQ3) to the integration of IoT in supply chain management and its associated challenges. Since 2018, there has been a noticeable increase in research output, signaling a growing academic interest in this area. Prominent

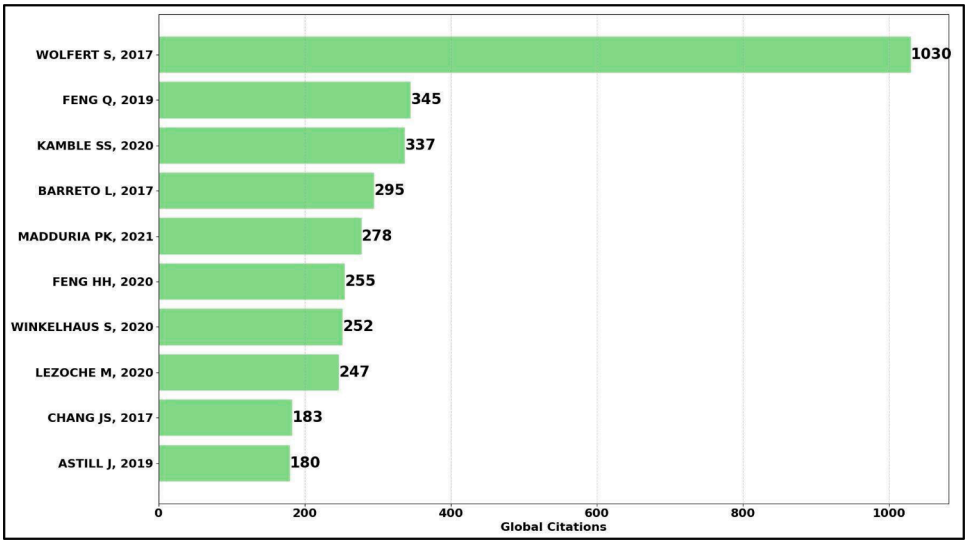


Fig. 5. Documents with most global citation per documents.

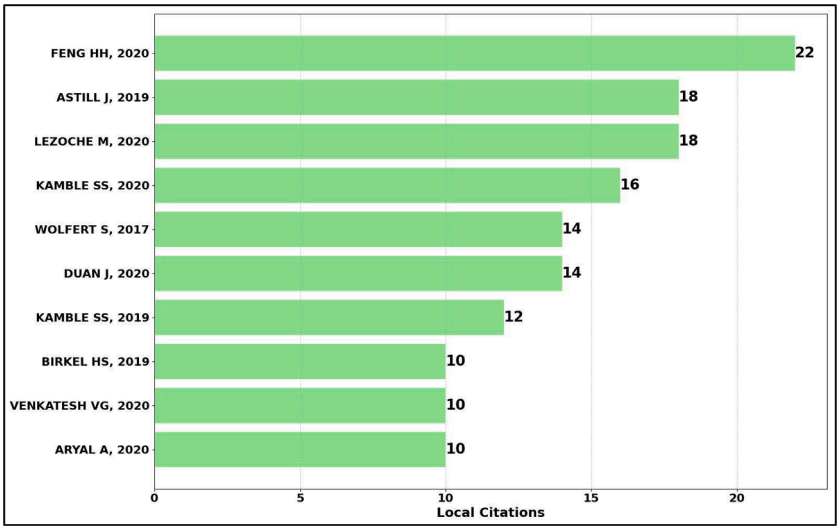


Fig. 6. Documents with most local citation per documents.

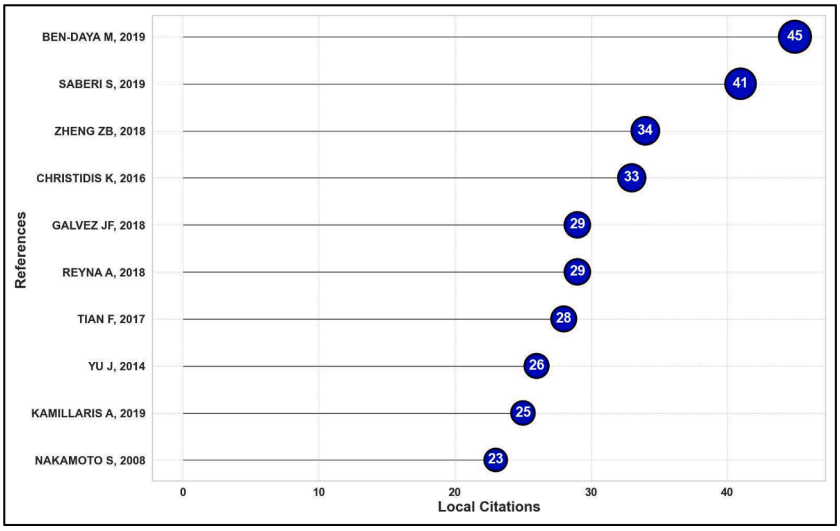


Fig. 7. Most local cited references.

journals such as Sustainability, International Journal of Production Research, and International Journal of Production Economics play vital roles in disseminating crucial findings, underscoring the significance of IoT in enhancing sustainability and efficiency in supply chain operations. The emphasis in the most influential articles on digital agriculture, privacy, and the integration of blockchain with IoT for traceability underscores the importance of these topics within the field. Moreover, the results indicate that while developing countries are rapidly expanding their research efforts, developed countries are also maintaining a strong and steady interest in these topics. Collectively, these findings address the initial three research questions and outline the current state of the research area.

3.2. Trend topic analysis result

The trend topic chart maps the evolution of research topics over time, showing their frequency within the literature. It features a scatter plot with time on the horizontal axis and frequent terms on the vertical axis. Each bubble represents a topic's median occurrence year, sized by occurrence frequency. Gray bars show the distribution's first and third quartiles, highlighting the variability and shift in research interests over time.

The result of trend topic analysis reveals "adoption", "safety" and "prediction" as burgeoning focal points in IoT and supply chain management research. This analysis, as depicted in Fig.10, underscores the escalating interest in these areas, alongside persistent topic like management, security, future, performance, privacy, food supply change and temperature. The significance of "adoption" is

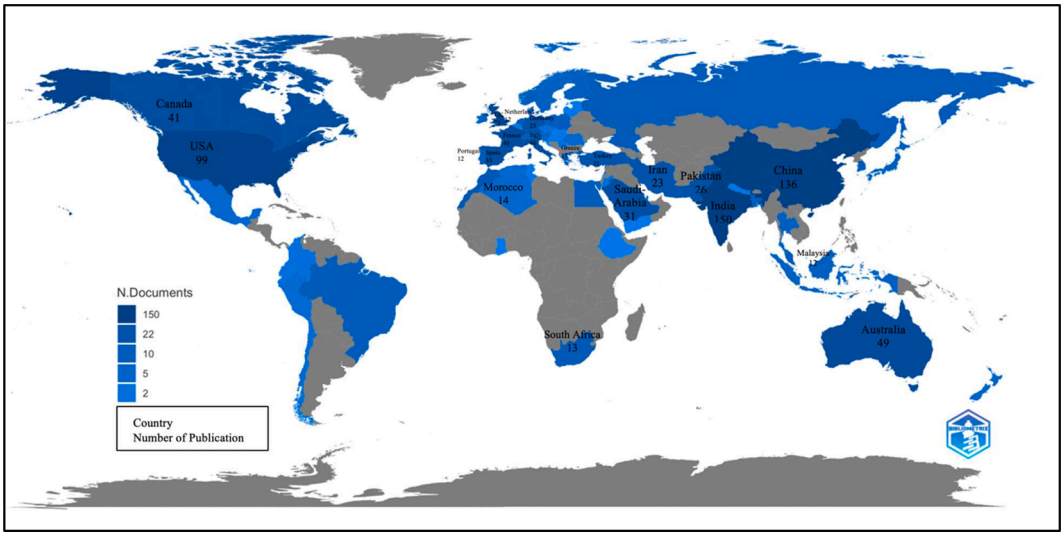


Fig. 8. Countries' publication status.

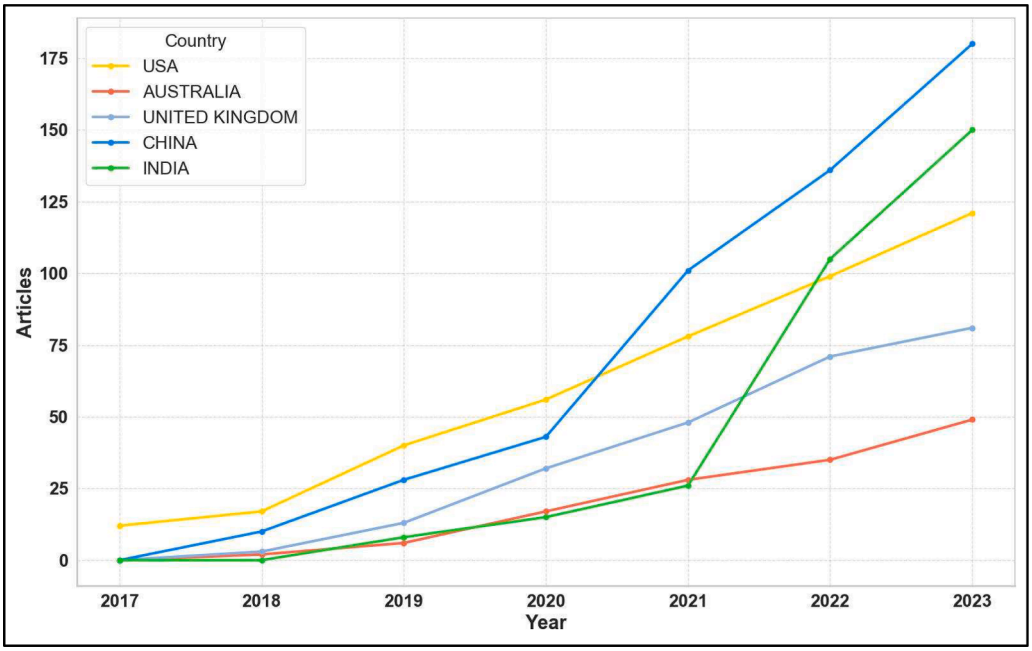


Fig. 9. Scientific production over time by country.

highlighted by its prominence at the top of the keyword list, indicating a key research emphasis in recent literature. The process of innovation adoption is delineated into three steps: "initiation", "adoption (decision)", and "implementation" [52]. In the context of this paper, the focus on adoption is pivotal as it sheds light on the integration of the IoT into the supply chain. Understanding the adoption process is crucial for comprehending the challenges and opportunities associated with incorporating IoT technologies. The examination of various studies elucidates a nuanced understanding of these challenges. For instance, Sharma et al., 2023 [53] address barriers and underscore the importance of overcoming challenges related to infrastructure expansion and digitalization. Lahane et al., 2023 [54] identify barriers on organizational, strategic, and social levels, emphasizing their substantial negative impact on the efficient adoption of Industry 4.0-enabled sustainable food supply chains. This study emphasizes the top management's perception and willingness to adopt these innovative practices as a critical barrier to successful integration. Highlighting challenges in BT-IoT adoption in the supply chain, Tavana et al., 2023 [55] underscore the importance of selecting the right platform and the supplier's ability to overcome barriers. Mardani & Saberi, 2023 [56] utilizes the q-rung fuzzy sets methodology to assess environmental, economic, and social drivers for adopting Industry 4.0 in the supply chain. A common thread across these studies is the acknowledgement

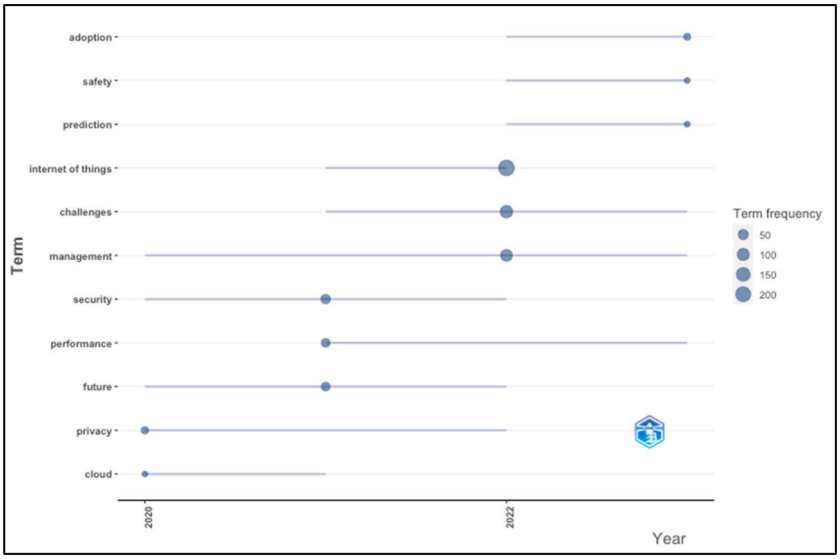


Fig. 10. Trend topic analysis.

of external hindrances, such as regulatory gaps and inadequate internet infrastructure, which pose significant challenges to IoT integration in supply chains including food supply chain [19].

Parallel to "adoption", the "safety" theme emerges as a critical concern in the result. A review on papers categorized by this topic reveals the topic is extensively used in context of food supply chains. Echegaray et al., 2022 [45] and Patel et al., 2023 [57] illustrate how Industry 4.0 technologies, including blockchain and IoT, are pivotal in enhancing food product traceability and safety, combating fraud, and minimizing waste. This collective inquiry underscores the transformative potential of IoT in fortifying food supply chains against safety risks and ensuring consumer health.

"Prediction" emerges as the third most prominent keyword according to the trend topic analysis. As depicted in Fig.10, this term first appeared in 2022, with a median publication frequency of approximately 50, indicating the topic's novelty and its growing appeal among researchers. In our study, "prediction" typically involves the application of advanced analytics, frequently enhanced by machine learning and artificial intelligence, to anticipate future conditions and outcomes in supply chain [58,59]. This technique is applied in various sectors that are applying IoT and big data, including healthcare supply chains, food supply chains and logistics

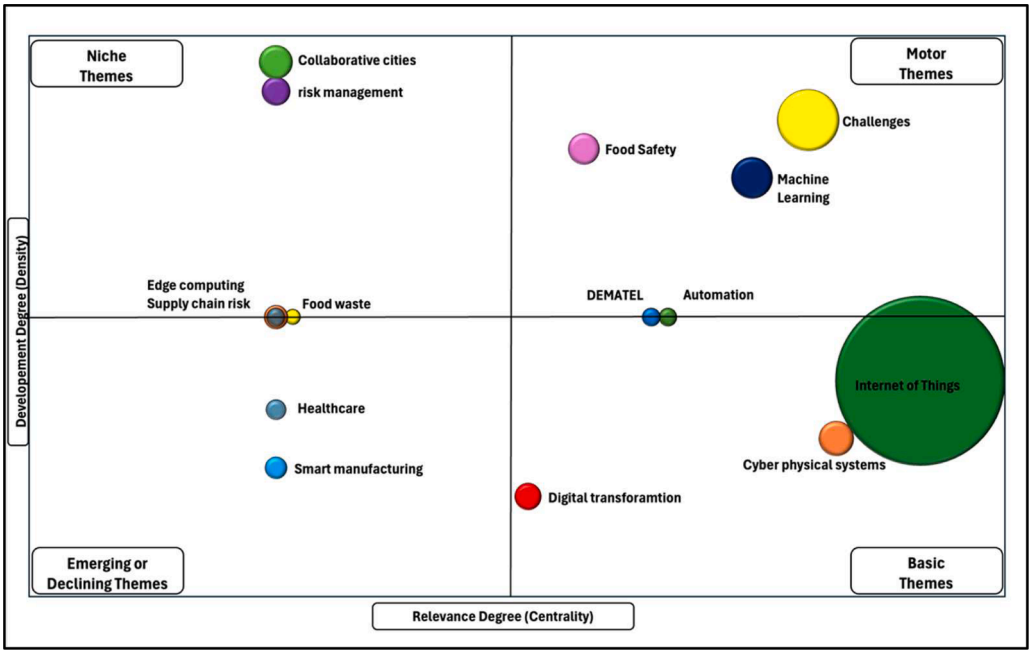


Fig. 11. Thematic analysis (first period).

[60–62].

In summary, the trend topic analysis responds to RQ4 by delineating the factors that have shaped the evolution of research at the intersection of IoT and supply chain challenges. The findings highlight the enduring significance of themes such as privacy, security, wireless sensor networks (WSN), and cloud computing, underscoring their fundamental role in the ongoing research into IoT integration with supply chains. These themes represent well-established areas of focus that continue to impact the development of secure and efficient IoT frameworks. Concurrently, emerging themes like IoT adoption in supply chains, the enhancement of food safety, and the application of predictive analytics are gaining popularity, indicating potential areas for future research. These newer themes suggest a shift towards developing more resilient and proactive supply chain management strategies.

3.3. Conceptual analysis result: thematic evolution

The conceptual result of research is framed as the result of thematic evolution of publications, in two periods 2009–2021, and 2022–2023 illustraged in Figs. 11 and 12. The four themes are assessed using Callon centrality and density. The trajectory of the evolution is analyzed, based on the clusters shown in two figures. The size of the bubbles shows the frequency of research focused on the theme.

3.3.1. Niche themes

Niche themes are the issues that are highly developed in literature, but not extremely research relevant [28]. As Fig.11, shows two themes including collaborative cities and risk management are highlighted as niche themes in the first period. The niche themes in the second period are knowledge graph, smart farming, digitalization, collaboration, food waste and authentication. Table 1 summarizes the details of niche themes.

3.3.2. Motor themes

The research on motor themes is very pertinent and extensively explored in the study. They have high levels of both centrality and density [28]. This theme in the first period covers three clusters including challenges, machine learning and food safety. In the second period, the machine learning cluster is again a motor theme. Ethereum, security and sustainability are also three other motor themes in this period. Table 2 provides a detailed summary of all motor themes, except for challenges, which forms the foundational basis of the research.

3.3.3. Basic themes

The basic thematic map shows the topics that despite being poorly researched, are categorized as important themes. Low density and high centrality are characteristics of the clusters that make up this quadrant [28]. Internet of things, cyber-physical systems, and digital transformation are detected as the main basic themes in the first period. Internet of things, digital transformation, digital twin, challenges, covid-19, reverse logistics, and circular economy are the basic themes in the second period. Table 3 provides a detailed summary of all basic themes, except for internet of things and challenges, which forms the foundational basis of the research.

3.3.4. Emerging or declining themes

The last quadrant displays the themes that are emerging or declining. The themes in this quadrant are made up of less significant and underdeveloped research concepts, and they appear at either the beginning or the end of the life cycle of research [28]. "Smart

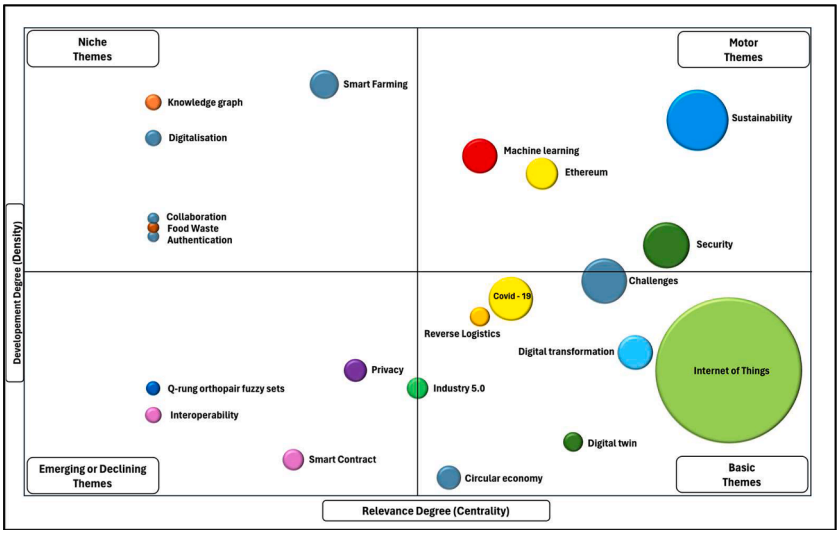


Fig. 12. Thematic analysis (second period).

Table 1
The clusters in niche themes.

Period	Cluster	Keywords	Interpretation	References
2009–2021	Collaborative cities	Food delivery, Material requirements planning theory, Post-harvest loss prevention, Smart Container, Smart home	-Facilitating the stakeholders' collaboration urban supply chain using IoT causes waste management, and resource sharing.	[63]
	Risk management	Risk management, Supply chain disruption, Supply risk	- Methods and frameworks for recognizing, evaluating, and reducing risks associated with IoT technology in supply chain operations.	[64,65]
2022–2023	Smart farming	Smart farming, Cloud server, Lora, Lorawan, Node red, Wireless monitoring	- Role of smart farming in enhancing the resilience of the agricultural supply chain by providing real-time data connectivity and traceability. - Highlights digital transformation in agriculture	[66–68]
	Authentication	Authentication	- The focus on developing and implementing advanced authentication methods to overcome the risks of cyber threats, safeguarding data integrity, ensuring system reliability, and maintaining seamless operations across different domains within the IoT-based supply chain	[69,70]
	Food waste	Food waste	- Focus on the growing importance of IoT in tackling the issue of food waste within the supply chain, supporting environmental sustainability, enhancing economic efficiency and operational effectiveness in the food industry.	[71,72]
	Digitalization	Digitalization, Safety	- Focusing on IoT, as a key technology in digitalization, that enhances sophisticated risk management strategies in supply chains. It provides real-time data and predictive insights to optimize operations, improve product quality, and ensure compliance with regulations.	[45,73]
	Knowledge Graph	Knowledge graphs, Production logistics	- IoT data populates the knowledge graph, addressing spatial and temporal disorders in the supply chain and ensuring data accuracy and real-time resource status, thus enhancing resource allocation precision.	[74]
	Collaboration	Collaboration	- Underscores the necessity for interconnected systems where stakeholders can cooperate efficiently to manage complexities and improve resilience and performance in an increasingly digitalized world	[60,75]

Table 2
The clusters in motor themes.

Period	Cluster	Keywords	Interpretation	References
2009–2021	Machine Learning	Machine learning, Artificial intelligence, Covid-19, Deep learning, Robotics	- Serves as a fundamental catalyst, adapting to challenges and integrating with AI, robotics, and information systems.	[76–78]
	Food Safety	Food safety, Food sustainability, Food quality, Traceability system	- Highlights IoT's role in real-time monitoring, ensuring product quality, and promoting sustainability in the food supply chain.	[79]
2022–2023	Machine Learning	Framework, Information system, Intrusion detection system, Supply chain performance, Supply chain resilience	- The ability of machine learning to process large volumes of data from diverse sources in real-time and make predictive assessments has become integral to addressing various challenges in IoT-enabled supply chains, including security, maintenance, operational resilience, and traceability	[80–83]
	Ethereum	Scalability, Performance, Hyper-ledger fabric	- Optimizes transaction speed and interoperability, resolving supply chain issues.	[84,85]
	Sustainability	Sustainability- Artificial intelligence (ai)- Big data - Smart city -Data analysis - Food waste reduction- Government-Resilience-Sensors - Smart cities	- Underscore the multidimensional role of IoT in creating a sustainable supply chain.	[86–89]
	Security	5 G, 6 G, cybersecurity, Edge computing, Applications, Smart logistics, Cloud	- The potential for security vulnerabilities due to increasing the integration of IoT with advanced digital technologies in supply chain, the expansion of complex and interconnected system architectures, and the critical need to safeguard these systems against a growing range of cybersecurity threats	[61, 90–93]

manufacturing" and "healthcare" stand out as developing clusters in the first period, garnering more attention due to their potential for transformation. Q-rung orthopair fuzzy sets, interoperability, privacy, and smart contract are all placed in the emerging or declining themes. Table 4 summarizes the details of each cluster in the emerging or declining themes.

Table 3
The clusters in basic themes.

Period	Cluster	Keywords	Interpretation	References
2009–2021	Digital transformation	Digital transformation	- Impact of digital transformation including AI and IoT in developing intelligent supply chain. - Highlighting the potential of digital tools to enhance productivity and manage supply chain risks more effective.	[94,95]
	Cyber-Physical Systems	Logistics 4.0, Risk assessment	- Focuses on enhancing supply chain visibility, automation, and efficiency through convergence of physical and digital elements.	[21, 96–98]
2022–2023	Digital transformation	Digital transformation, Automation, Precision agriculture, Smart sensors, Virtualization	- Highlights digital transformation in supply chain and role IoT as a digital technology for solving global crises such as covid-19, climate change, sustainability, and efficiency, resilience -Pointing out the challenges in digital technology adoption in supply chain	[99–102]
	Digital Twin	Digital twins	- The synergy of IoT and digital twin in context of supply chain provide real-time data capture, enhanced security, operational simulation, and strategic decision-making support. - - Visibility, improved efficiency, and greater resilience in managing supply chain challenges	[62,103]
	Circular Economy	Circular supply chain, Conceptual framework	- Emphasizes sustainability through resource reuse, waste reduction, and lifecycle management in IoT-enabled supply chains.	[104–107]
	Covid-19	Covid-19, Logistics 4, healthcare, RFID, Simulation, Smart Supply Chain, Vaccine Supply Chain	- IoT's role in enhancing resilience and overcoming challenges such as demand fluctuation in supply chain during Covid-19. -The ability of technology in improving vaccine and test-kit distribution.	[108–112]
	Reverse logistics	Reverse logistics, Localization	-Deployment of the IoT sensors to capture the production real-time information for transparency and accuracy in reverse logistics and its potential to enhance efficiency and environment sustainability	[113–115]

3.3.5. Border themes

Based on Fig.11 five themes appeared on the border of themes in the first period. The food waste, supply chain risk and edge computing are three themes that have emerged on the border of niche themes and emergent themes. The DEMATEL and automation themes are placed on the border of basic and motor themes. Industry 5.0 illustrated in Fig.12 is the only border theme, in the second period and is located on the border of emergent and basic theme. Table 5 provides a detailed summary of all border themes.

The presence of sustainability in motor theme in the second period shows a clear movement towards integrating sustainability into IoT-enhanced supply chains [61,89] . The recent emphasis on themes like sustainability and circular economy highlights a shift toward environmental consciousness in supply chain management [16,111]. IoT technologies are used to reduce environmental impacts, such as waste and carbon footprint, reflecting a broader trend towards more sustainable operations for environmental responsibility [140].

The thematic progression also suggests a deeper digital transformation within supply chains. Themes like digital twin and digitalization signal the adoption of sophisticated digital models and simulations, allowing for more accurate predictions, better risk management, and enhanced decision-making capabilities. Digital twins, in particular, serve as a bridge between physical and digital worlds, offering real-time monitoring and simulation capabilities that enhance both efficiency and resilience [103].

The emergence of themes related to authentication and security reflects the importance of cybersecurity. As IoT devices proliferate across supply chains, ensuring the security of these devices and the data they handle becomes paramount. The research indicates a

Table 4
The clusters in emerging or declining themes.

Period	Cluster	Keywords	Interpretation	References
2009–2021	Smart manufacturing	Smart manufacturing, Smart	- Focuses on how IoT technologies drive data-driven innovations in smart manufacturing, integrating physical and virtual elements. Addressing cybersecurity and government challenges for IoT advancements in the realm of smart manufacturing.	[116–118]
	Healthcare	Healthcare	- Emphasizes on IoT ability to provide safe, reliable, and efficient solution at a very detailed and specific level from different health parameters.	[119–121]
2022–2023	Interoperability	Interoperability	- Addresses challenges related to converging systems and technologies, facilitating seamless data exchange in IoT-enabled supply chain activities.	[122–124]
	q-rung orthopair fuzzy sets	q-rung orthopair fuzzy sets	- Investigates q-rung fuzzy sets to manage uncertainty in decision-making within IoT-enabled supply chain contexts, providing reliable decision outputs.	[125–127]
	Privacy	Privacy, Attacks	- Privacy as a vital factor in developing dependable and trustworthy IoT applications - The importance of exploring technologies which protect personal data and improves privacy	[128,129]
	Smart contract	Smart contract	-Improving efficiency and transparency and automation in supply chain by integrating IoT and blockchain-based smart contracts.	[130–132]

Table 5
The clusters on borders.

Period	Cluster	Keywords	Interpretation	
2009–2021	Food Waste	Food waste	- Aims to reduce food waste in supply chains using IoT and RFID technology for monitoring and optimization.	[79]
	Supply Chain Risk	Supply chain risk	- Highlights the role of IoT in addressing supply chain risks. IoT, along with online optimization, assesses and mitigates risks in real-time.	[133,134]
	Edge Computing	Edge Computing, Interoperability	Highlighting the role of edge computing in addressing operational efficiency, security, and data management challenges in using IoT in supply chain.	[135–137]
	Automation	Automation	- Focuses on integrating robotics and smart sensors to automate and optimize various supply chain functions.	[138]
	DEMATEL	DEMATEL	- Essential analytical tool for pinpointing and ranking elements in complex supply chain and IoT interactions.	[19,134]
2022–2023	Industry 5.0	Industry 5.0	- Focuses on collaboration between humans and machines, introducing advanced technologies for efficient manufacturing solutions.	[139]

growing awareness and implementation of cybersecurity measures to protect against potential cyber threats and data breaches [69,70,91].

The integration of advanced technologies such as machine learning, blockchain-based Ethereum, and edge computing within IoT frameworks is another key point. These technologies not only enhance existing capabilities [80,85,137] but also open new avenues for innovation within supply chains, such as blockchain's potential to enhance transparency and traceability [46,51].

The thematic evolution underscores an increase in cross-sector collaboration facilitated by IoT. As supply chains become more complex and intertwined, the ability to seamlessly integrate and collaborate across different sectors and geographical boundaries is enhanced, fostering innovation through shared resources and capabilities [60,75].

Analyzing the thematic evolution in IoT and supply chain management reveals a significant expansion in focus. By comparing papers from the first period (2009–2021) to those from the second period (2022–2023), it is clear that the initial use of IoT to enhance operational efficiency in supply chains has expanded to also applying IoT for improving resilience. This expansion is evident in the evolution of research themes from 2009 to 2023.

Initially, the integration of IoT in supply chains focused on enhancing efficiency, with prominent themes like edge computing minimizing latency and enhancing response times by processing data near its source, thereby optimizing operational efficiencies [17]. Machine learning and automation simplified various supply chain functions, increasing efficiency, and reducing human error through smart sensors and robotics [138].

IoT's role in real-time monitoring also ensured product quality and sustainability, particularly in themes like food safety and smart manufacturing, where real-time data enabled immediate quality control measures essential for maintaining standards and regulatory compliance [116,141].

However, the approach of studies in the second period represents a significant turn towards enhancing resilience within supply chains, influenced by global disruptions such as the COVID-19 pandemic [2,60,102]. This newer phase introduces themes like "Smart Farming," emphasizing IoT's role in increasing the resilience of agricultural supply chains through real-time data connectivity and traceability [67], and "Authentication," addressing the need for robust cybersecurity measures [70]. "Digitalization and Digital Twins" highlight IoT's support for advanced risk management strategies and decision-making processes through technologies like Digital Twins, optimizing operations and enhancing responsiveness and recovery during disruptions [62,70].

The impact of the pandemic, particularly noted in the COVID-19 theme, has significantly shaped recent research, highlighting IoT's role in managing supply chain resilience during crises [83,142]. IoT technologies have been pivotal in adapting to demand fluctuations and enhancing the logistics of distributing critical supplies like vaccines [143].

This shift from a focus on efficiency to resilience in IoT-enabled supply chains represents a crucial reorientation in response to global supply chain disruptions. While the initial drive towards efficiency aimed at streamlining processes and reducing costs, the challenges posed by the COVID-19 pandemic and other factors have underscored the necessity for supply chains that are not only efficient but also resilient, adaptable, and secure. The current emphasis on resilience prepares supply chains to better withstand, adapt to, and recover from disruptions. This involves using IoT not just for operational efficiency but also integrating it with cybersecurity measures, real-time data analytics, and advanced decision-support systems to form resilient, adaptable, and sustainable supply chain networks. This shift in study topics reflects a wider understanding of the necessity for supply chains to achieve a balance between flexibility and efficiency to handle upcoming obstacles and disruptions and make sure they can survive and prosper in the face of uncertainty across the world.

In summary, the thematic evolution from 2009 to 2021 focused on enhancing operational efficiencies through technologies like edge computing, machine learning, and automation, ensuring product quality and sustainability through real-time monitoring. Transitioning into the period from 2022 to 2023, the focus expands to include resilience, driven by global disruptions such as the COVID-19 pandemic. Themes in this period highlight IoT's evolving role in enhancing supply chain resilience and securing data integrity, supporting advanced risk management strategies, and enhancing decision-making processes. This evolution suggests an ongoing commitment to integrating IoT within supply chains to enhance both efficiency and resilience against future disruptions.

In response to RQ4, technological advancements and a heightened focus on security and sustainability due to global challenges have driven the research towards developing more resilient and adaptive supply chain systems. Future trajectories are expected to

continue emphasizing resilience, with advanced analytics and industry 4.0 technologies such as digital twins playing pivotal roles in predictive and responsive strategies. These developments suggest an ongoing commitment to integrating IoT within supply chains, not only to enhance efficiency but also to ensure resilience and robustness against future disruptions, reflecting a deeper, more thorough approach toward modern supply chain management in an increasingly digitalized world.

3.4. Network analysis result

A network analysis primarily illustrates the cooperative ties between IoT-supply chain publications. Keyword co-occurrence and co-citation network analyses are shown and analyzed in this part.

3.4.1. Keyword co-occurrence

The keyword co-occurrence network map using VoSviewer is shown in Fig. 13, with each circle denoting a keyword and its size indicating how frequently it occurs. The research clusters of keywords are represented by the color of the circles, and the degree of connection and similarity between the keywords is shown by the distance between the circles. Based on this map, the Internet of Things (178 co-occurrences) is the most significant node, followed by blockchain technology (137), supply chain (85), industry 4.0 (62), sustainability (30), security (29), big data (21), food supply chain (19). The map shows three significant clusters in green, red, and blue colors related to the central concepts of supply chain and IoT. Each cluster represents a unique aggregation of themes and concerns within the realm of our study. Further analysis of each cluster reveals more detailed insights into their respective themes and contributions to the study, as discussed below.

In the green cluster, shared keywords like “big data”, “artificial intelligence”, “machine-learning”, “cybersecurity” and “agriculture” highlights the significance of integrating digital technologies with IoT and the cybersecurity concerns in this regard. The integration of advanced technologies including bid data, artificial intelligence, and machine learning with IoT technology enables data-driven insights, enhancing operational efficiency and providing real-time visibility into supply chain processes. Furthermore, the emphasis on “IoT security” and “intrusion detection system” underscores a commitment to ensuring the security of interconnected systems. Kumar & Mallipeddi,2022 [144] considered recent progress in information and communication technologies, including IoT to be instrumental in fostering the implementation of Industry 4.0 and Industry 5.0 in operational and supply chain management. They emphasize the cybersecurity risk as a significant hurdle for organizations in development that employ and integrate IoT with other technologies. Aliahmadi & Nozari, 2023 [145] also count the security concern in the integration of IoT and blockchain as a subset of cybersecurity and propose a fuzzy-neutrosophic method to analyze the related indicators. Addressing cybersecurity challenges necessitates a robust cybersecurity framework, safeguarding against data threats, device vulnerabilities, and supply chain attacks. Prioritizing data privacy and managing integration challenges in cloud computing is crucial. Streamlining cybersecurity, enhancing data protection, and fostering adaptability form a unified approach for both domains, acknowledging their interconnected nature and the imperative to navigate challenges collaboratively. Within the red cluster, “Internet of Things” and “blockchain” and “supply chain”

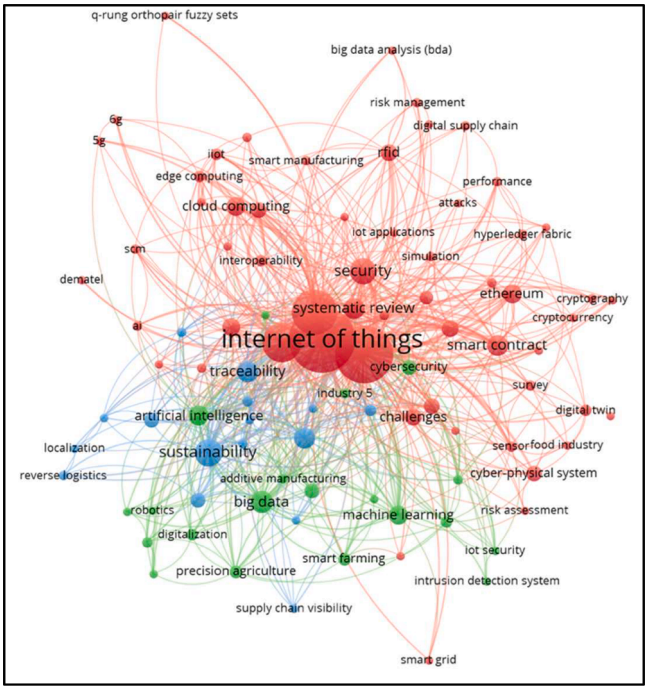


Fig. 13. Author’s keyword co-occurrence network.

appear as high-occurrence words. Notably, terms such as “smart contract”, “security”, “logistics”, “RFID” and “Ethereum” stand out as high-frequency keywords shared between both domains. This word set illustrates the synergies between IoT and blockchain in the supply chain, highlighting how their intersection can significantly enhance security measures within logistics and supply chain operations. Utilizing innovative technologies tied with blockchain technology such as smart contracts, RFID, and Ethereum, not only fortifies security but also introduces efficiency and transparency. The inclusion of “risk assessment” indicates a focus on evaluating potential vulnerabilities within the integrated system. These keywords collectively embody a vision of a technologically advanced and secure supply chain ecosystem, emphasizing the transformative impact of integrating IoT and supply chain practices through innovative applications and blockchain technology.

The high-frequency co-occurrence of keywords in blue clusters like “sustainability”, “traceability”, “food supply chain”, “circular economy” and “digital transformation” in previous research involving the IoT and supply chain reflects a comprehensive approach to optimizing the efficiency of supply chain considering the environmental sustainability. The emphasis on sustainability signals a commitment to environmentally conscious practices, aligning with broader environmental goals. Additionally, the focus on addressing food supply chain underscores the application of IoT and data analytics to minimize inefficiencies and enhance traceability. Overall, this co-occurrence suggests an integrated approach to applying IoT and advanced data analytics for enhancing traceability and transparency in a sustainable and efficient supply chain with a specific focus on food supply chain.

Collectively, the three main clusters—green, red, and blue—result from keyword co-occurrence, each highlighting a unique issue to response RQ4. The green cluster focuses on technology integration, such as AI within IoT and cybersecurity concerns. The red cluster explores the synergies between IoT and blockchain, enhancing both security and efficiency. Meanwhile, the blue cluster emphasizes sustainability and traceability, particularly in food supply chains, advocating for environmentally conscious practices. Based on the result, the integration of IoT within supply chain operations is driven by the need for enhanced traceability, efficiency, environmental sustainability, and security. Technological advancements such as AI, blockchain, and big data analytics serve as key drivers by enabling more precise monitoring and decision-making. However, barriers include significant cybersecurity risks, the complexity of IoT implementation across diverse supply chain components, and the substantial investment required in technology and training. Additionally, the resistance to change within organizations and the need for robust data privacy measures also pose challenges to IoT integration.

3.4.2. Co-citation results

The co-citation network analysis depicted in Fig. 14. In this visualization, the size of the labels and circles for each reference depends on the number of citations that item has received. Specifically, items that have been cited more frequently are depicted with larger labels and circles. The proximity between two references in the visualization indicates a stronger relationship between them. The most significant co-citation connections between references are additionally illustrated with lines. Fig.14 illustrates three clusters at the intersection of IoT, supply chains, and their challenges, organized according to a co-citation analysis. The green cluster represents papers that review the challenges of integrating IoT with other technologies to enhance supply chain operations and the competitive advantages of IoT in the supply chain. The blue cluster focuses on the role of blockchain in addressing deficiencies and privacy challenges that may arise from using IoT technology within the supply chain. While the green and blue cluster focuses on the technology-related issues, the red cluster is more focused on using IoT-blockchain technology for improving the agri-food sector supply chain. The papers in this cluster also assess barriers to integrating these technologies into supply chain operations.

The papers in the green cluster examine how the IoT ecosystem affects the supply chain. They discuss its transformative impact, various effects, and the challenges organizations face when integrating it. Gubbi et al., 2013 [146] underscore critical elements such as cloud software architectures and pervasive communication networks, which require attention to realize the IoT vision. Additionally,

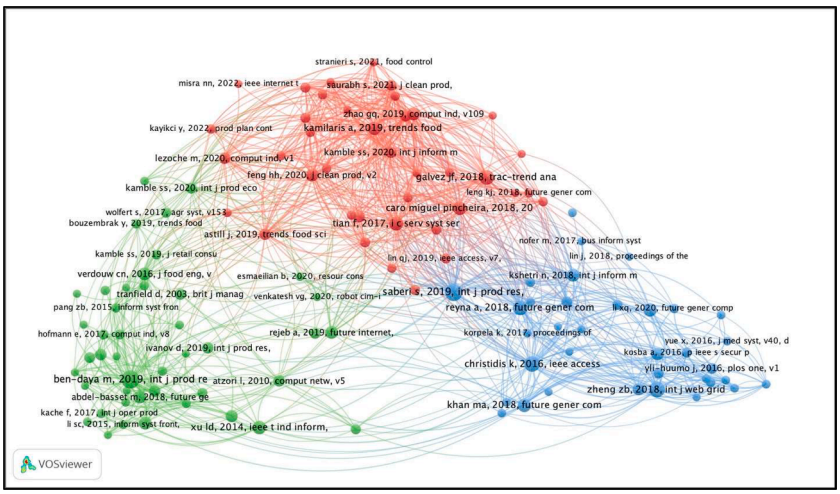


Fig. 14. Reference co-citation network.

the study acknowledges analytics tools as another vital technological requirement. Al-Fuqaha et al., 2015 [147] identify standardization and interoperability as hindrances to achieving effective interaction and cooperation among different protocols. The paper also explains how these technological challenges could disrupt achieving horizontal integration, which ensures seamless collaboration among diverse IoT services in the supply chain. Some other technological barriers, including security and privacy concerns, scalability and manageability challenges, energy efficiency and power consumption issues data management and analytics challenges, and integration with other emerging technologies such as cloud and fog computing are also mentioned and elaborated on in this study. Miorandi et al., 2012 [148] propose a systematic methodology for comprehending challenges associated with IoT in the supply chain. This framework identifies potential technological barriers such as reliability, real-time processing, and energy efficiency. The study goes beyond a narrow technology-centric perspective to ensure successful IoT implementation and emphasizes additional considerations like cost-effectiveness, user acceptance, and ethical implications. Xu et al., 2014 [149] emphasize the importance of radio frequency identification (RFID), wireless sensor networks (WSN), near-field communication (NFC), and big data analytics as enabling technology in an IoT service-oriented architecture. The paper also highlights how integrating these technologies with IoT can improve the service level in the healthcare, manufacturing, transportation, agriculture, and energy industries. Verdouw et al., 2016 [150] address the challenges of adopting IoT in the supply chain particularly, for small and medium-sized enterprises (SMEs) that may not have the necessary resources and expertise. The study highlights the adoption costs as the cost of hardware, software, and infrastructure. The paper proposes an architecture that supports food companies to easily connect to virtual objects in a secure and trusted way while reducing the cost of adoption and improving the return on investment (ROI). Ben-Daya et al., 2022b [6] utilize a resource-based conceptual framework to figure out how internal supply chain skills such as collaboration, innovation, integration, quality, and visibility are strengthened by IoT resources to address external challenges and improve supply chain management competitiveness such as resilience, reliability, traceability, and velocity. Khan & Salah, 2018 [151] sheds light on the benefits of IoT in the supply chain including real-time monitoring and tracking of items, which improves the administration and operations of the supply chain. Tzounis et al., 2017 [152] highlight the challenges of IoT in agriculture, including hardware and software obstacles involving the development of durable, cost-effective devices and efficient data processing algorithms. Interoperability hurdles related to standardizing data formats and protocols for seamless communication among diverse devices and systems is the other issue. Networking is also a challenge that encompasses the need for robust communication networks capable of managing substantial data loads in remote areas. Security concerns involve safeguarding communication, data privacy, and defense against cyber threats in distributed IoT systems. Lastly, stack challenges pertain to creating a versatile and scalable architecture accommodating various devices, applications, and services while remaining adaptable to changing environments. Atzori et al., 2010 [153] identify IoT security challenges and categorize them into three groups of low, intermediate, and high-level problems. It proposes robust blockchain-based remedies as a suitable solution. Moreover, the paper emphasizes the capability of blockchain in facilitating decentralized management, governance, and tracking across the IoT device supply chain, thereby addressing concerns regarding trust, authentication, and data security for all stakeholders.

The blue color cluster highlights how adding blockchain can be beneficial for creating a competitive advantage in to supply chain and decreasing the risks associated with adopting IoT. This category delves into solutions that improve trust and data security while examining the dynamic interplay between IoT security and blockchain integration. The papers also review the critical issues and challenges for the integration of these technologies including standardization, scalability, and security. Zheng et al., 2018 [154] highlight three key issues with blockchain including scalability due to limited transaction processing in current systems, privacy issues arising from the transparency of blockchain, and the need to prevent self-centered mining for network security. This study emphasizes the benefits of blockchain in IoT, such as direct coin and sensor data exchange in distributed autonomous companies. It also underscores how blockchain might improve IoT security, privacy, and provenance, suggesting a privacy-preserving method for commissioning IoT devices and mentioning IBM's ADEPT (Autonomous Decentralized P2P Telemetry) system, which uses blockchain to create a self-managing network of devices. Kshetri, 2018 [155] underlines the significance of integrating IoT with blockchain solutions to verify the identities of individuals and assets, enhancing accountability and transparency in supply chain operations, optimizing cost, quality, speed, reliability, risk mitigation, sustainability, and flexibility. Reyna et al., 2018 [156] emphasize that despite the disruptive applications of IoT and blockchain in the supply chain, challenges persist, including issues like a lack of standardization, scalability concerns, limitations in device capabilities, security considerations, and the complexity of applications that require attention and resolution. Saberi et al., 2019 [157] enumerates enhancing efficiency as the advantages of blockchain adoption in smart contracts. This paper highlights streamlined procedures and reduced intermediary involvement, as well as greater transparency, traceability, and security are the main contribution of smart contract enabled by blockchain in the supply chain operation. Kosba et al., 2016 [158] discusses on the privacy issues in financial transaction on the blockchain and introduced Hawk, as a decentralized system and IdealP cash to address this concern. Abeyratne & Monfared, 2016 [159] enumerate cultural and technical barriers such as standardization, interoperability, and scalability as factors hindering the integration of blockchain. Conversely, it highlights the potential opportunities this technology offers to improve security, transparency, traceability, efficiency, and reduce costs, and foster collaboration among stakeholders in the manufacturing supply chain. Casino et al., 2019 [160] emphasizes on automation as an advantage enabled by integrating blockchain IoT devices connected to cryptocurrency-based accounts for microtransactions.

The red cluster focuses on using advanced technologies, like blockchain and IoT, to improve the agri-food sector's supply chain management. Traceability in the supply chain is one common point, meticulously discussed in the papers in this cluster. Tian, 2016 [161] proposes an agri-food supply chain traceability system that leverages RFID and blockchain technology to enhance data authenticity and real-time tracking across various stages, aiming to improve efficiency and reduce losses in logistics. Galvez et al., 2018 [162] focus on the stakeholder perspective, highlighting how blockchain-enhanced traceability from farm to table can benefit a wide

range of participants in the supply chain by increasing transparency, efficiency, and security. Christidis & Devetsikiotis, 2016 [163] introduce the limitations of IoT-based traceability systems reliant on centralized frameworks, as data integrity, tampering, and vulnerabilities, and point out the incorporation of fully decentralized traceability blockchain technology within the Agri-Block-IoT system, offers a robust resolution for this challenge. This integration ensures the safeguarding of data integrity, resistance to tampering, and the facilitation of comprehensive transaction traceability. Zhao et al., 2019 [164] assess the challenges and opportunities for the agri-food value chain. It uncovers the potential of blockchain in enhancing agri-food performance, focusing on areas like food safety, quality, and traceability. The study highlights potential applications, including product traceability, information security, manufacturing optimization, and sustainable water management. However, challenges arise, encompassing storage scalability, privacy issues, excessive costs, throughput, and skill gaps. Lezoche et al., 2020 [78] introduce the concept of "Agri-Food 4.0″ which adapts the principles of industry 4.0 to the agricultural domain, aiming to enhance supply chains through digital technologies including blockchain, IoT, big data analytics, and AI. The study highlights adding real-time crop monitoring, automated management, enhanced efficiency, reduced costs, improved environmental practices, heightened customer satisfaction, novel business models, and advanced machine connectivity to the benefits. The paper also adjoins sectoral diversity, investment constraints, technical skill shortages, privacy concerns, data security, rural connectivity limitations, and the intricacies of decentralization to previous challenges. Kamilaris et al., 2019 [165] use a maturity-level framework from conceptual design to integration into regular operations to assess the challenges of IoT-blockchain adoption in the food supply chain. Policy and regulatory aspects are the added challenges in this paper. The paper also highlights the importance of evaluating the economic sustainability of current initiatives in the food supply chain. H. Feng et al., 2020b [51] explain that conventional IoT traceability systems, highlight their limitations stemming from centralized structures that impede consumers' comprehensive access to transaction details and product origins. It proposes a blockchain-based traceability architecture to improve food traceability. The study introduces not only benefits, such as efficiency but also organizational and social challenges that exist in this context. Kamble et al., 2020 [166] suggest a mathematical model to prioritize the most important enabler for blockchain adoption in the agricultural supply chain. The study employs interpretive structural modeling and decision-making trial and evaluation laboratory (DEMATEL) techniques to understand the complex connections among blockchain technology enablers. Traceability emerges as the most crucial driver for blockchain technology adoption in agricultural supply chain. The research establishes relationships between blockchain technology adoption factors, highlighting traceability, auditability, immutability, and provenance as significant. Enablers encompass features like privacy, decentralized database, smart contracts, and transparency. Challenges encompass organizational, technological, and environmental aspects. Salah et al., 2019 [167] introduce a blockchain-based solution using Ethereum smart contracts to enhance soybean traceability in the agricultural supply chain. This approach eliminates the need for a central authority, ensuring transparency, security, and efficiency. The solution highlights the role of smart contracts in governing interactions among supply chain stakeholders, recording transactions on an immutable blockchain ledger with connections to a decentralized file system.

Based on co-citation analysis IoT provides the real-time monitoring feature and tracking capability in the supply chain that can enhance the efficiency of supply chain operation. Applying IoT in the supply chain may require enabling technologies like RFID, WSNs, NFC, and big data analytics. The decentralized feature of blockchain technology makes its integration to IoT an efficient strategy to increase trust, authentication, governance, and tracking capabilities. It facilitates the verification of identities for individuals and assets, ensuring transparency and traceability across parameters such as cost, quality, speed, reliability, risk mitigation, sustainability, and flexibility. By streamlining procedures and reducing intermediary involvement, blockchain promotes enhanced efficiency, lower costs, and improved collaboration among stakeholders. The technology's inherent accountability safeguards transactions, secure sensitive data, including medical records, and streamlines overall business operations. In the context of IoT operations, blockchain ensures secure data sharing, improves efficiency, and automates processes using RFID sensors. Furthermore, the integration of cryptocurrency-based accounts improves transaction processes, adding layer of security and efficiency to financial transactions within the supply chain.

Despite all the benefits of IoT and blockchain for the supply chain, some challenges yet exist and need to be addressed to enhance the efficiency of applying these solutions in the supply chain. Interoperability issues and standardization challenges cast shadows on the seamless integration of IoT with its required technologies like big data analytics, cloud, and fog computing. The scalability and manageability of IoT systems present hurdles, accompanied by energy efficiency and power consumption considerations. Data management complexities, encompassing the collection and sharing of vast datasets, coupled with analytics challenges, underscore the intricate nature of IoT adoption. Integration with other emergent technologies like cloud and fog computing further complicates the landscape. Beyond technological barriers, the challenges extend to ensuring reliability, real-time processing capabilities, cost-effectiveness, and user acceptance. Ethical considerations weave through the fabric of IoT implementation, while the overarching costs, spanning hardware, software, and infrastructure, pose substantial barriers to entry.

The broad landscape of IoT, introduces new vulnerabilities and potential threats and raises the security concerns. While blockchain adds a layer of security, it does not eliminate all risks associated with IoT devices. Interoperability, standardization, and scalability issues are also shared challenges for adopting both technologies. Both technologies involve managing vast amounts of data, creating concerns about the secure collection, sharing, and analysis of this information. Integration with other emerging technologies, including cloud and fog computing, poses common hurdles related to compatibility and collaboration. Additionally, cultural, and technical barriers are challenges applicable to both IoT and blockchain implementation. IoT challenges also encompass energy efficiency, power consumption, and user acceptance. Self-centered mining and limited device capabilities are also issues tied to integrating blockchain.

The merits of integration of IoT and blockchain in the agri-food sector supply chain is the evidence of their effect in all industries' supply chains. These technologies hold immense potential to revolutionize efficiency and sustainability by establishing a transparent and secure traceability framework for agri-food products. This transformative synergy offers a range of benefits, including improved

security measures, stringent quality management, and the assurance of enhanced consumer trust in the origin and quality of agri-food items. The use of blockchain introduces features such as auditability, immutability, and provenance, which play a significant role in establishing the authenticity and integrity of the supply chain. Notably, the application of Ethereum smart contracts within the blockchain ecosystem further contributes to enhancing traceability, automating processes, and ensuring the seamless execution of agreements throughout the agri-food supply chain. The combined strengths of IoT and blockchain technologies in this context promise to create a more resilient, accountable, and consumer-centric agri-food supply chain. The challenges in adopting IoT and blockchain technologies in the agri-food sector supply chain mirror the broader challenges observed in their diverse application. In the agricultural context, technical challenges persist, specifically the ones related to the development of durable and cost-effective IoT devices and the formulation of efficient data processing algorithms, highlighting the sector's unique requirements. Organizational and interoperability hurdles emphasize the need for standardizing data formats and protocols for seamless communication, underscoring the sector-specific complexities in coordinating diverse devices and systems. Networking challenges arise from the necessity for robust communication networks capable of handling substantial data loads, particularly in remote agricultural areas, further emphasizing the sector's specific connectivity challenges. Security concerns, despite the added layer of security from blockchain, remain a prominent issue, necessitating measures to safeguard communication channels and fortify defenses against cyber threats in distributed IoT systems within the agriculture domain. The economic sustainability of agricultural initiatives, regulatory and policy considerations, as well as environmental sustainability, are highlighted as overarching challenges, specifying the unique intricacies of the agri-food supply chain.

In summary, the co-citation analysis categorizes the research into three thematic clusters—green, red, and blue— to answer RQ5. The green cluster delves into the complexities and transformative effects of IoT, emphasizing technology integration challenges like cybersecurity and standardization, crucial for enhancing operational efficiencies. The red cluster explores the intersection of IoT and blockchain technologies, highlighting their potential to improve agri-food supply chain management through enhanced traceability and security. The blue cluster discusses the role of blockchain in mitigating privacy and security concerns within IoT-enabled supply chains, advocating for improved transparency and data security. Together, these clusters illustrate the evolving landscape of IoT in supply chains, addressing both technological advancements and the persistent challenges of security, interoperability, and integration. The co-citation analysis reveals that IoT plays a pivotal role in enhancing real-time monitoring and tracking within supply chains, thereby improving operational efficiencies. The integration of blockchain with IoT emerges as a strategic approach to augment trust, governance, and tracking while addressing security and privacy challenges, which are critical for adopting IoT in supply chains. Challenges persist, particularly around interoperability, scalability, and standardization, which affect the seamless integration of IoT

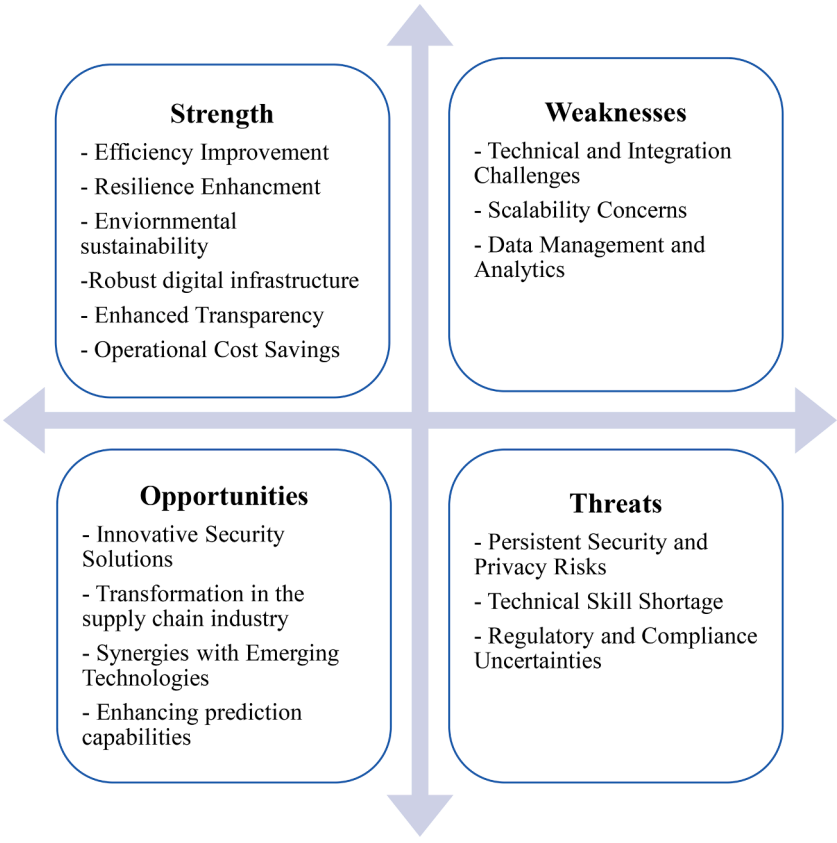


Fig. 15. SWOT matrix of IoT in supply chain.

and its related technologies such as big data analytics, cloud, and fog computing. Moreover, the complexities of data management and the need for reliable real-time processing underline the intricate nature of IoT adoption. These technological and organizational challenges highlight the necessity for a robust framework that supports both IoT and blockchain technologies to enhance the security and efficiency of supply chain operations.

4. Discussion

Recent years have witnessed a significant surge in research exploring the challenges in intersection of the Internet of Things and supply chain management, reflecting an escalating interest in understanding the complexities associated with this nexus on a global scale. This increase in scholarly attention can stem from the expanding deployment of IoT technologies within supply chain, aimed at enhancing efficiency, sustainability and resilience and mitigating uncertainties prevalent in supply chain processes.

In harmony with this global perspective, in our comprehensive investigation into the challenges of integration of IoT within supply chains, we structured our analysis around specific research questions that progressively built upon each other to provide a thorough view of the field. Initially, we employed descriptive analysis to address RQ1, RQ2, and RQ3, laying a solid foundation by mapping out the publication trends, identifying key papers, and highlighting geographical contributions that inform the current state of IoT in supply chains. These insights were essential in setting an environment for the analysis of deeper issues.

Progressing to RQ4, we utilized trend topic and conceptual analyses to delve deeper into the thematic evolution within the literature, uncovering how research topics have shifted and expanded over time. This analysis shed light on the innovative aspects of IoT applications in supply chains, identifying emerging themes that signify shifts in technology adoption and application strategies.

For RQ5, our focus turned to network analysis, which included exploring keyword co-occurrence and co-citation networks. This methodological approach was instrumental in revealing the interconnected nature of the themes and how they influence one another, thereby providing a more nuanced understanding of the ecosystem within which IoT operates in supply chains.

Building upon these layered insights, we conducted a SWOT analysis to identify the strengths, weaknesses, opportunities, and threats at the intersection of IoT and supply chains. This analysis served as the synthesis of our findings, drawing directly from the detailed explorations of each research question to pinpoint the strategic elements that stakeholders need to consider. Fig. 15 illustrates the SWOT matrix, with subsequent sections providing detailed insights into each identified factor.

4.1. Strength

“Efficiency improvement” is the first strength of applying IoT in supply chain. IoT’s real-time monitoring and tracking capabilities are unmatched, significantly increasing the supply chain’s efficiency, responsiveness, and adaptability. These strengths are critical in fast-paced market environments where timely information can drastically improve decision-making and operational agility. “Resilience enhancement” in supply chains, particularly in the context of global challenges like the COVID-19 pandemic is one of the most important strengths of the integration of IoT in supply chain. IoT technology plays a crucial role in providing real-time data connectivity and traceability, thereby strengthening the resilience of different supply chains. This capability is critical for responding effectively to disruptions, ensuring continuity, and maintaining the stability of supply chain operations during crises and uncertainties. “Environmental sustainability” is significantly propelled by the integration of IoT technologies in to supply chain that contribute to reducing environmental impacts such as waste and carbon footprint. “Robust digital infrastructure” can also fortify and increase motivations for applying IoT in supply chain. The employment of RFID, WSNs, NFC, and big data analytics, along with blockchain, creates a strong and resilient technological ecosystem. This integration not only enhances data accuracy and security but also establishes a framework for trust and authentication among all supply chain stakeholders, ensuring every transaction and movement is verifiable. The other factor that is highlighted in this study as a strength is “Enhanced transparency across the supply chain”. The combination of IoT and blockchain technologies ensures transparency and traceability. This is particularly crucial in managing the supply chain’s complexity, offering clear visibility into cost, quality, speed, reliability, and sustainability metrics. Such transparency is invaluable in risk mitigation and enhancing flexibility within supply chain operations. “Operational cost savings” is another advantage of applying IoT and industry 4.0 technologies in businesses. Streamlining processes and minimizing the need for intermediaries through IoT can lead to significant operational cost savings. This efficiency gain, coupled with improved stakeholder collaboration, underscores a strategic advantage in adopting these technologies.

4.2. Weaknesses

“Technical and integration challenges” is the first factor identified as the weakness of IoT in supply chain. Despite its potential, the seamless integration of IoT technology within the supply chain is encounter challenges, including interoperability issues and the absence of universal standards. These technical barriers can hinder the broad adoption of IoT solutions. “Scalability concerns” have also arisen challenges for integrating IoT in to supply chain. As supply chains grow in complexity and scope, the scalability and manageability of IoT systems, especially concerning energy efficiency and power consumption, emerge as critical challenges. These issues necessitate innovative solutions to ensure IoT infrastructure can adapt and scale without compromising performance. “Data Management and analytics” and the complexity of handling vast datasets generated by IoT devices is a significant challenge. This complexity is amplified by the need for sophisticated analytics capabilities to derive actionable insights from data, further complicated by the integration of cloud and fog computing technologies.

4.3. Opportunities

Although data security is one of the most prevalent threats in the literature, “Innovative security solutions” is an opportunity that can solve this issue. The integration of blockchain offers novel solutions to longstanding security and privacy concerns within the IoT ecosystem. By providing a secure, decentralized framework for data management, blockchain technology can significantly mitigate risks associated with data breaches and unauthorised access. “Transformation in the supply chain industry” and the sector-specific application highlights the broader applicability of IoT and blockchain across various industries, promising a new era of supply chain management. The application of IoT and blockchain within the supply chain industry specifically agri-food supply chain exemplifies the potential for these technologies to revolutionize industry standards for efficiency, sustainability, and consumer trust. “Synergies with emerging technologies”, or the convergence of IoT with cutting-edge developments in Industry 4.0, artificial intelligence, and cybersecurity presents unique opportunities for enhancing supply chain operations. These synergies underscore the potential for IoT to function as a catalyst in the digital transformation of supply chains, fostering innovation and operational excellence. "Enhancing prediction capabilities in supply chain" underlines the adoption of advanced analytics, machine learning, and artificial intelligence to forecast future conditions and optimizing decision-making across various sectors. Particularly in sectors like healthcare, logistics, and food supply chains, enhanced predictive capabilities through IoT can lead to more proactive management, better resource allocation, and reduced inefficiencies.

4.4. Threats

“Persistent security and privacy risks” is the first and most important concern highly observed in the literature. Despite the advantages offered by blockchain, the IoT landscape remains vulnerable to security threats. These vulnerabilities necessitate continuous vigilance and innovation in cybersecurity measures to protect against evolving threats. The implementation of IoT and blockchain in supply chains encounters various adoption barriers, including cultural resistance, technical complexities, and the need for substantial initial investments. Overcoming these barriers requires concerted efforts in education, training, and demonstrating the clear value proposition of these technologies. “Regulatory and compliance uncertainties” can also disrupt the effective application of IoT in supply chain. The dynamic regulatory landscape poses additional challenges for IoT and blockchain adoption in supply chains. Navigating these uncertainties requires an initiative-taking approach to compliance and engagement with regulatory developments to ensure IoT implementations are both effective and legally sound. “Technical skill shortage” is one of the challenges identified as a threat in the literature. Organizations may find it difficult to innovate and successfully implement new technology if they lack this kind of experience. To overcome these obstacles and fully reap the benefits of IoT systems, it is necessary to make sure that staff members are knowledgeable in these fields

5. Conclusion, limitation, and future research

The overarching goal of our research was to outline the current challenges and future trajectories of IoT in supply chains, with a specific focus on understanding the drivers and barriers affecting its adoption. To achieve this, we conducted a comprehensive bibliometric analysis. We analyzed 408 publications, selected from a total of 913, that were published from 2009 to 2023. This analysis answered five research questions, revealing current publication trends, the most influential articles, the top contributing countries, the evolution and future trajectories, and the drivers and barriers in the context of IoT and supply chain. Building on our research findings, we conducted a SWOT analysis to gain a deeper understanding of the IoT-based supply chain ecosystem.

In terms of academic output, we observed a significant increase in studies exploring the challenges at the intersection of IoT and supply chain management starting from 2018. This surge in research is globally distributed, with substantial contributions from both developed and developing countries, including notable publications in Sustainability journal. Furthermore, themes such as privacy and security have been thoroughly discussed in the present research. Emerging topics in current studies include the use of prediction models in supply chain by integrating IoT and machine learning, food safety, and decisions regarding IoT adoption in supply chains. The results of the study show that although initial research on the role of IoT in supply chains focused on enhancing reliability and operational efficiency, the emphasis has shifted post-COVID-19. Recent studies are now exploring the impact of IoT on enhancing supply chain resilience, going beyond mere efficiency improvements. We also notice a recent focus on sustainability, circular economy, digital transformation and Ethereum as trending themes in IoT-based supply chain. In terms of drivers and barriers of IoT in supply chain, IoT’s role in enhancing supply chain efficiency, resilience, and sustainability is evident. The technology facilitates real-time tracking and optimization of logistics and inventory management, especially within the agri-food and production sectors. It enables better decision-making through the continuous generation and monitoring of data, which is crucial for adapting to changing business conditions. Security and privacy concerns, interoperability issues, and the lack of standardization are significant barriers. Additionally, the scalability of IoT systems, energy efficiency, and the high initial costs of infrastructure and skill acquisition pose further obstacles. The result shows blockchain technology has been identified as a promising solution to many of these challenges, particularly in enhancing cybersecurity and facilitating decentralized trust. Blockchain can improve data integrity and transparency, which are critical for the seamless operation of IoT-enabled supply chains. We conducted a SWOT analysis to comprehensively outline all the factors identified as strengths, opportunities, weaknesses, and threats associated with integrating IoT in supply chain management. The proposed solutions and insights can help organizations navigate the complexities of IoT adoption and contribute to seamless integration of IoT within the supply chain landscape.

Even though our study contributes insights into the integration of IoT in supply chains, it does have some limitations that could be

considered further in future studies. Our reliance on secondary data from existing literature and the WoS database may have excluded pivotal works available in other databases such as Scopus, Google Scholar, or PubMed. Specific research clusters within IoT and supply chain integration, such as industry-specific challenges or regional variations, might have been overlooked. Moreover, the evolving nature of IoT technology and supply chain requirements means that our findings may quickly become outdated as new technologies, methodologies and necessities emerge. In addition, it is important to consider the inherent biases and limitations of the analytical tools used, such as Biblioshiny and VOSviewer. Future studies may benefit from a more comprehensive database selection, and continuous updates to reflect the latest developments in the field.

Enhancing supply chain resilience through IoT is currently a focal point in research. However, the presence of COVID-19 in the theme with low density of study, indicates core scope for further investigation. Additionally, the integration of IoT with digital technologies such as digital twins, its impact on the circular economy, and its role in Industry 5.0 and reverse logistics are critical areas that have not been sufficiently explored. These represent significant opportunities for future research.

CRediT authorship contribution statement

Atefeh Shoomal: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammad Jahanbakht:** Writing – review & editing, Validation, Supervision, Project administration, Conceptualization. **Paul J. Componation:** Supervision. **Dervis Ozay:** Writing – review & editing, Software, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the data on Mendeley data

References

[1] S.K. Paul, P. Chowdhury, A production recovery plan in manufacturing supply chains for a high-demand item during COVID-19, Int. J. Phys. Distrib. Logist. Manag. 51 (2021) 104–125, https://doi.org/10.1108/IJPDLM-04-2020-0127.
[2] K. Govindan, S.P. Sethi, T.C.E. Cheng, S.F. Lu, Designing supply chain strategies against epidemic outbreaks such as COVID-19: Review and future research directions, Decis. Sci. 54 (2023) 365–374, https://doi.org/10.1111/deci.12609.
[3] M. Materikina, A. Shoomal, L.H. Manh, Y. Zhou, Investigating Production Yield Effect on Inventory Control through a Hybrid Simulation Approach, in: 2023 Winter Simulation Conference (WSC), IEEE, 2023, pp. 2088–2099, https://doi.org/10.1109/WSC60868.2023.10407418.
[4] F. Dabbene, P. Gay, C. Tortia, Traceability issues in food supply chain management: A review, Biosyst. Eng. 120 (2014) 65–80, https://doi.org/10.1016/j.biosystemseng.2013.09.006.
[5] B. Adenso-Diaz, C. Mena, S. García-Carbajal, M. Liechty, The impact of supply network characteristics on reliability, Supply Chain Manag. 17 (2012) 263–276, https://doi.org/10.1108/13598541211227108.
[6] M. Ben-Daya, E. Hassini, Z. Bahroun, A conceptual framework for understanding the impact of internet of things on supply chain management, Oper. Supply Chain Manag. 15 (2022) 251–268, https://doi.org/10.31387/oscm0490345.
[7] D. Ivanov, A. Dolgui, B. Sokolov, The impact of digital technology and Industry 4.0 on the ripple effect and supply chain risk analytics, Int. J. Prod. Res. 57 (2019) 829–846, https://doi.org/10.1080/00207543.2018.1488086.
[8] M.E. Porter, J.E. Heppelmann, How smart, Harv. Bus. Rev. Retrieved from (2014). https://hbr.org/2014/11/how-smart-connected-products-are-transforming-competition.
[9] V.V. Prabhakar, C.S. Belarmin Xavier, K.M. Abubeker, A Review On Challenges and Solutions in the Implementation of Ai, IoT and Blockchain in Construction Industry, Mater Today Proc, 2023, https://doi.org/10.1016/j.matpr.2023.03.535.
[10] R.M.L. Rebelo, S.C.F. Pereira, M.M. Queiroz, The interplay between the Internet of things and supply chain management: challenges and opportunities based on a systematic literature review, Benchmarking 29 (2022) 683–711, https://doi.org/10.1108/BIJ-02-2021-0085.
[11] F. Huang, J. Chen, L. Sun, Y. Zhang, S. Yao, Value-based contract for smart operation and maintenance service based on equitable entropy, Int. J. Prod. Res. 58 (2020) 1271–1284, https://doi.org/10.1080/00207543.2019.1617450.
[12] F. Bienhaus, A. Haddud, Procurement 4.0: factors influencing the digitisation of procurement and supply chains, Bus. Process Manag. J. 24 (2018) 965–984, https://doi.org/10.1108/BPMJ-06-2017-0139.
[13] T. de Vass, H. Shee, S.J. Miah, IoT in supply chain management: a narrative on retail sector sustainability, Int. J. Logist. Res. Appl. 24 (2021) 605–624, https://doi.org/10.1080/13675567.2020.1787970.
[14] R. Agrifoglio, C. Cannavale, E. Laurenza, C. Metallo, How emerging digital technologies affect operations management through co-creation. Empirical evidence from the maritime industry, Prod. Plan. Control. 28 (2017) 1298–1306, https://doi.org/10.1080/09537287.2017.1375150.
[15] G.J. Hahn, Industry 4.0: A supply chain innovation perspective, Int. J. Prod. Res. 58 (2020) 1425–1441, https://doi.org/10.1080/00207543.2019.1641642.
[16] V. Varriale, A. Cammarano, F. Michelino, M. Caputo, Sustainable supply chains with blockchain, IoT and RFID: A simulation on order management, Sustainability. 13 (2021) 6372–6395, https://doi.org/10.3390/su13116372.
[17] I. Bosi, J. Rosso, E. Ferrera, C. Pastrone, IIot Platform for Agile Manufacturing in Plastic and Rubber Domain, in: Proceedings of the 5th International Conference on Internet of Things, Big Data and Security, SCITEPRESS - Science and Technology Publications, 2020, pp. 436–444, https://doi.org/10.5220/0009573304360444.
[18] T. de Vass, H. Shee, S. Miah, IoT in supply chain management: opportunities and challenges for businesses in early industry 4.0 context, Operations and supply chain management, An Int. J. 14 (2021) 148–161, https://doi.org/10.31387/oscm0450293.
[19] S.S. Kamble, A. Gunasekaran, H. Parekh, S. Joshi, Modeling the internet of things adoption barriers in food retail supply chains, J. Retail. Consum. Serv. 48 (2019) 154–168, https://doi.org/10.1016/j.jretconser.2019.02.020.

[20] A. Rejeb, J.G. Keogh, H. Treiblmaier, Leveraging the internet of things and blockchain technology in supply chain management, Future Internet. 11 (2019) 161–183, https://doi.org/10.3390/fi11070161.
[21] T. Sobb, B. Turnbull, N. Moustafa, Supply chain 4.0: A survey of cyber security challenges, solutions and future directions, Electronics. (Basel) 9 (2020) 1–31, https://doi.org/10.3390/electronics9111864.
[22] S. Ding, H. Ward, S. Cucurachi, A. Tukker, Revealing the hidden potentials of internet of things (IoT) - An integrated approach using agent-based modelling and system dynamics to assess sustainable supply chain performance, J. Clean. Prod. 421 (2023) 138558–138574, https://doi.org/10.1016/j.jclepro.2023.138558.
[23] I. Expósito, J.A. Gay-Fernández, I. Cuiñas, R. Maxwell, A complete traceability system for a wine supply Chain using radio-frequency identification and wireless sensor networks, IEEe Antennas. Propag. Mag. 55 (2013) 255–267, https://doi.org/10.1109/MAP.2013.6529365.
[24] A. Shoomal, Challenges of IoT-enabled supply chain, Mendeley Data, v1, 2024, https://doi.org/10.17632/8fwr3ckpjk.1.
[25] M. Aria, C. Cuccurullo, bibliometrix: an R-tool for comprehensive science mapping analysis, J. Informetr. 11 (2017) 959–975, https://doi.org/10.1016/J.JOI.2017.08.007.
[26] D. Ozay, M. Jahanbakht, P.J. Componation, A. Shoomal, State of the Art and Themes of the Research on Artificial Intelligence (AI) Integrated Customer Relationship Management (CRM): Bibliometric Analysis and Topic Modelling, in: 2023 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD), IEEE, 2023, https://doi.org/10.1109/ICTMOD59086.2023.10438124.
[27] D. Ozay, M. Jahanbakht, A. Shoomal, S. Wang, Artificial intelligence (AI)-based customer relationship management (CRM): a comprehensive bibliometrics and systematic literature review with outlook on future research, Enterp. Inf. Syst. 18 (2024) 1–40, https://doi.org/10.1080/17517575.2024.2351869.
[28] A. Belfiore, C. Cuccurullo, M. Aria, IoT in healthcare: A scientometric analysis, Technol. Forecast. Soc. Change 184 (2022) 1–10, https://doi.org/10.1016/j.techfore.2022.122001.
[29] K. Börner, C. Chen, K.W. Boyack, Visualizing knowledge domains, Annual Rev. Inf. Sci. Tech. Rev. 37 (2003) 179–255, https://doi.org/10.1002/aris.1440370106.
[30] M. Callon, J.P. Courtial, F. Laville, Co-word analysis as a tool for describing the network of interactions between basic and technological research: The case of polymer chemistry, Scientometrics. 22 (1991) 155–205, https://doi.org/10.1007/BF02019280.
[31] M.J. Cobo, A.G. López-Herrera, E. Herrera-Viedma, F. Herrera, An approach for detecting, quantifying, and visualizing the evolution of a research field: a practical application to the fuzzy sets theory field, J. Informetr. 5 (2011) 146–166, https://doi.org/10.1016/j.joi.2010.10.002.
[32] J.P. Courtial, A coword analysis of scientometrics, Scientometrics. 31 (1994) 251–260, https://doi.org/10.1007/BF02016875.
[33] N. Coulter, I. Monarch, S. Konda, Software engineering as seen through its research literature: A study in co-word analysis, J. Am. Soc. Inf. Sci. 49 (1998) 1206–1223, https://doi.org/10.1002/(SICI)1097-4571(1998)49:13<1206::AID−ASI7>3.0.CO;2-F.
[34] T. Cahlik, Comparison of the maps of science, Scientometrics. 49 (2000) 373–387, https://doi.org/10.1023/A:1010581421990.
[35] M. Aria, M. Misuraca, M. Spano, Mapping the evolution of social research and data science on 30 years of social indicators research, Soc. Indic. Res. 149 (2020) 803–831, https://doi.org/10.1007/s11205-020-02281-3.
[36] H.N. Su, C. Lee P., Mapping knowledge structure by keyword co-occurrence: a first look at journal papers in technology foresight, Scientometrics. 85 (2010) 65–79, https://doi.org/10.1007/s11192-010-0259-8.
[37] H. Small, Co-citation in the scientific literature: A new measure of the relationship between two documents, J. Am. Soc. Inf. Sci. 24 (1973) 265–269, https://doi.org/10.1002/asi.4630240406.
[38] M. Hussain, W. Javed, O. Hakeem, A. Yousafzai, A. Younas, M.J. Awan, H. Nobanee, A.M. Zain, Blockchain-based IoT devices in supply chain management: a systematic literature review, Sustainability. 13 (2021) 13646–13669, https://doi.org/10.3390/su132413646.
[39] J. Nurgazina, U. Pakdeetrakulwong, T. Moser, G. Reiner, Distributed ledger technology applications in food supply chains: A review of challenges and future research directions, Sustainability. 13 (2021) 4206–4232, https://doi.org/10.3390/su13084206.
[40] S. Ahmed, T. Kalsoom, N. Ramzan, Z. Pervez, M. Azmat, B. Zeb, M. Ur Rehman, Towards supply chain visibility using internet of things: A dyadic analysis review, Sensors 21 (2021) 4158–4182, https://doi.org/10.3390/s21124158.
[41] R. Singh, A.D. Dwivedi, G. Srivastava, Internet of things based blockchain for temperature monitoring and counterfeit pharmaceutical prevention, Sensors 20 (2020) 3951–3974, https://doi.org/10.3390/s20143951.
[42] M.N.M. Bhutta, A.A. Khwaja, A. Nadeem, H.F. Ahmad, M.K. Khan, M.A. Hanif, H. Song, M. Alshamari, Y. Cao, A survey on blockchain technology: evolution, architecture and security, IEEe Access. 9 (2021) 61048–61073, https://doi.org/10.1109/ACCESS.2021.3072849.
[43] Z. Yu, S.A.R. Khan, M. Mathew, M. Umar, M. Hassan, M.J. Sajid, Identifying and analyzing the barriers of internet of things in sustainable supply chain through newly proposed spherical fuzzy geometric mean, Comput. Ind. Eng. 169 (2022) 108227–108240, https://doi.org/10.1016/j.cie.2022.108227.
[44] V.S. Yadav, A.R. Singh, R.D. Raut, S.K. Mangla, S. Luthra, A. Kumar, Exploring the application of Industry 4.0 technologies in the agricultural food supply chain: A systematic literature review, Comput. Ind. Eng. 169 (2022) 108304–108323, https://doi.org/10.1016/J.CIE.2022.108304.
[45] N. Echegaray, A. Hassoun, S. Jagtap, M. Tetteh-Caesar, M. Kumar, I. Tomasevic, G. Goksen, J.M. Lorenzo, Meat 4.0: principles and applications of industry 4.0 technologies in the meat industry, Appl. Sci. 12 (2022) 6986–7005, https://doi.org/10.3390/app12146986.
[46] W. Alkhader, N. Alkaabi, K. Salah, R. Jayaraman, J. Arshad, M. Omar, Blockchain-based traceability and management for additive manufacturing, IEEe Access. 8 (2020) 188363–188377, https://doi.org/10.1109/ACCESS.2020.3031536.
[47] R. Alfred, J.H. Obit, C.P.-Y. Chin, H. Haviluddin, Y. Lim, Towards paddy rice smart farming: a review on big data, machine learning, and rice production tasks, IEEe Access. 9 (2021) 50358–50380, https://doi.org/10.1109/ACCESS.2021.3069449.
[48] A. Al Sadawi, M.S. Hassan, M. Ndiaye, A survey on the integration of blockchain with IoT to enhance performance and eliminate challenges, IEEe Access. 9 (2021) 54478–54497, https://doi.org/10.1109/ACCESS.2021.3070555.
[49] S. Wolfert, L. Ge, C. Verdouw, M.J. Bogaardt, Big data in smart farming – A review, Agric Syst 153 (2017) 69–80, https://doi.org/10.1016/J.AGSY.2017.01.023.
[50] Q. Feng, D. He, S. Zeadally, M.K. Khan, N. Kumar, A survey on privacy protection in blockchain system, J. Netw. Comput. Appl. 126 (2019) 45–58, https://doi.org/10.1016/J.JNCA.2018.10.020.
[51] H. Feng, X. Wang, Y. Duan, J. Zhang, X. Zhang, Applying blockchain technology to improve agri-food traceability: A review of development methods, benefits and challenges, J. Clean. Prod. 260 (2020) 121031–121046, https://doi.org/10.1016/J.JCLEPRO.2020.121031.
[52] J.L. Pierce, A.L. Delbecq, Organization structure, individual attitudes and innovation, Acad. Manage. Rev. 2 (1977) 27–39, https://doi.org/10.2307/257602.
[53] A. Sharma, A. Sharma, R.K. Singh, T. Bhatia, Blockchain adoption in agri-food supply chain management: an empirical study of the main drivers using extended UTAUT, Bus. Process Manag. J. 29 (2023) 737–756, https://doi.org/10.1108/BPMJ-10-2022-0543.
[54] S. Lahane, V. Paliwal, R. Kant, Evaluation and ranking of solutions to overcome the barriers of Industry 4.0 enabled sustainable food supply chain adoption, Clean. Logist. Supply Chain. 8 (2023) 100116–100134, https://doi.org/10.1016/j.clscn.2023.100116.
[55] M. Tavana, A. Khalili Nasr, A.B. Ahmadabadi, A.S. Amiri, H. Mina, An interval multi-criteria decision-making model for evaluating blockchain-IoT technology in supply chain networks, Internet of Things 22 (2023) 100786–100810, https://doi.org/10.1016/j.iot.2023.100786.
[56] A. Mardani, S. Saberi, Industry 4.0 adoption drivers for sustainable supply chain in the manufacturing sector using a hybrid decision-making approach under q-rung orthopair fuzzy information, IEEe Trans. Eng. Manage (2024) 1–18, https://doi.org/10.1109/TEM.2023.3262961.
[57] A.S. Patel, M.N. Brahmbhatt, A.R. Bariya, J.B. Nayak, V.K. Singh, Blockchain technology in food safety and traceability concern to livestock products, Heliyon. 9 (2023) e16526, https://doi.org/10.1016/j.heliyon.2023.e16526.
[58] L. Ren, Z. Jia, Y. Laili, D. Huang, Deep learning for time-series prediction in IIoT: progress, challenges, and prospects, IEEe Trans. Neural Netw. Learn. Syst. (2024) 1–20, https://doi.org/10.1109/TNNLS.2023.3291371.
[59] G. Dong, M. Tang, Z. Wang, J. Gao, S. Guo, L. Cai, R. Gutierrez, B. Campbel, L.E. Barnes, M. Boukhechba, Graph neural networks in IoT: a survey, ACM. Trans. Sens. Netw. 19 (2023) 1–50, https://doi.org/10.1145/3565973.

[60] P. Kumar, R.K. Singh, A. Shahgholian, Learnings from COVID-19 for managing humanitarian supply chains: systematic literature review and future research directions, Ann. Oper. Res. 335 (2024) 899–935, https://doi.org/10.1007/s10479-022-04753-w.

[61] S. Ahmadzadeh, T. Ajmal, R. Ramanathan, Y. Duan, A comprehensive review on food waste reduction based on IoT and big data technologies, Sustainability. 15 (2023) 3482–3501, https://doi.org/10.3390/su15043482.

[62] T. Zdolsek Draksler, M. Cimperman, M. Obrecht, Data-driven supply chain operations—The pilot case of postal logistics and the cross-border optimization potential, Sensors 23 (2023) 1624–1643, https://doi.org/10.3390/s23031624.

[63] D. Bogataj, M. Bogataj, D. Hudoklin, Mitigating risks of perishable products in the cyber-physical systems based on the extended MRP model, Int. J. Prod. Econ. 193 (2017) 51–62, https://doi.org/10.1016/j.ijpe.2017.06.028.

[64] M. Pishdar, F. Ghasemzadeh, J. Antucheviciene, J. Saparauskas, Internet of things and its challenges in supply chain management: A rough strength-relation analysis method, E a M: Ekonomie a Management 21 (2018) 208–222, https://doi.org/10.15240/tul/001/2018-2-014.

[65] H.S. Birkel, E. Hartmann, Impact of IoT challenges and risks for SCM, Supply Chain Manag. 24 (2019) 39–61, https://doi.org/10.1108/SCM-03-2018-0142.

[66] M. Saban, M. Bekkour, I. Amdaouch, J. El Gueri, B. Ait Ahmed, M.Z. Chaari, J. Ruiz-Alzola, A. Rosado-Muñoz, O. Aghzout, A smart agricultural system based on PLC and a cloud computing web application using LoRa and LoRaWan, Sensors 23 (2023) 2725–2741, https://doi.org/10.3390/s23052725.

[67] S. Shreya, K. Chatterjee, A. Singh, BFSF: A secure IoT based framework for smart farming using blockchain, Sustain. Comput.: Inform. Syst. 40 (2023) 100917–100933, https://doi.org/10.1016/j.suscom.2023.100917.

[68] M. Saban, O. Aghzout, A. Rosado-Munoz, Deployment of a LoRa-based Network and Web Monitoring Application for a Smart Farm, in: 2022 IEEE International Workshop on Metrology for Industry 4.0 & IoT (MetroInd4.0&IoT), IEEE, 2022, pp. 424–427, https://doi.org/10.1109/MetroInd4.0IoT54413.2022.9831521.

[69] X. Jiang, R. Dou, Q. He, X. Zhang, W. Dou, EdgeAuth: An intelligent token-based collaborative authentication scheme, Softw. Pract. Exp. (2023), https://doi.org/10.1002/spe.3206.

[70] F. Stodt, C. Reich, Bridge of Trust: Cross domain authentication for industrial internet of things (IIoT) blockchain over transport layer security (TLS), Electronics. (Basel) 12 (2023) 2401–2417, https://doi.org/10.3390/electronics12112401.

[71] R. Ramanathan, Y. Duan, T. Ajmal, K. Pelc, J. Gillespie, S. Ahmadzadeh, J. Condell, I. Hermens, U. Ramanathan, Motivations and challenges for food companies in using IoT sensors for reducing food waste: Some insights and a road map for the future, Sustainability. 15 (2023) 1665–1687, https://doi.org/10.3390/su15021665.

[72] A. Lamberty, J. Kreyenschmidt, Ambient parameter monitoring in fresh fruit and vegetable supply chains using internet of things-enabled sensor and communication technology, Foods. 11 (2022) 1777–1798, https://doi.org/10.3390/foods11121777.

[73] Z. Wei, H. Liu, X. Tao, K. Pan, R. Huang, W. Ji, J. Wang, Insights into the application of machine learning in industrial risk assessment: A bibliometric mapping analysis, Sustainability. 15 (2023) 6965–6995, https://doi.org/10.3390/su15086965.

[74] Z. Zhao, M. Zhang, J. Chen, T. Qu, G.Q. Huang, Digital twin-enabled dynamic spatial-temporal knowledge graph for production logistics resource allocation, Comput. Ind. Eng. 171 (2022) 108454–108467, https://doi.org/10.1016/j.cie.2022.108454.

[75] E. Pessot, A. Zangiacomi, R. Fornasiero, Unboxing the hyper-connected supply chain: a case study in the furniture industry, Prod. Plan. Control. 35 (2024) 580–598, https://doi.org/10.1080/09537287.2022.2110958.

[76] M. Attaran, Digital technology enablers and their implications for supply chain management, Supply Chain Forum 21 (2020) 158–172, https://doi.org/10.1080/16258312.2020.1751568.

[77] D.C. Nguyen, M. Ding, P.N. Pathirana, A. Seneviratne, Blockchain and AI-Based solutions to combat coronavirus (COVID-19)-like epidemics: A survey, IEEe Access. 9 (2021) 95730–95753, https://doi.org/10.1109/ACCESS.2021.3093633.

[78] M. Lezoche, H. Panetto, J. Kacprzyk, J.E. Hernandez, M.M.E. Alemany Díaz, Agri-food 4.0: A survey of the supply chains and technologies for the future agriculture, Comput. Ind. 117 (2020) 103187–103202, https://doi.org/10.1016/j.compind.2020.103187.

[79] R. Badia-Melis, U. Mc Carthy, L. Ruiz-Garcia, J. Garcia-Hierro, J.I. Robla Villalba, New trends in cold chain monitoring applications - A review, Food Control 86 (2018) 170–182, https://doi.org/10.1016/j.foodcont.2017.11.022.

[80] H.K. Skrodelis, A. Romanovs, Synthetic Network Traffic Generation in IoT Supply Chain Environment, in: 2022 63rd International Scientific Conference on Information Technology and Management Science of Riga Technical University (ITMS), IEEE, 2022, pp. 1–5, https://doi.org/10.1109/ITMS56974.2022.9937126.

[81] S.P. Gayialis, E.P. Kechagias, G.D. Konstantakopoulos, G.A. Papadopoulos, A predictive maintenance system for reverse supply chain operations, Logistics 6 (2022) 4–19, https://doi.org/10.3390/logistics6010004.

[82] A. Yazdinejad, E. Rabieinejad, T. Hasani, G. Srivastava, A BERT-based recommender system for secure blockchain-based cyber physical drug supply chain management, Cluster. Comput. 26 (2023) 3389–3403, https://doi.org/10.1007/s10586-023-04088-6.

[83] T. Ahmed, C.L. Karmaker, S.B. Nasir, M.A. Moktadir, S.K. Paul, Modeling the artificial intelligence-based imperatives of industry 5.0 towards resilient supply chains: A post-COVID-19 pandemic perspective, Comput. Ind. Eng. 177 (2023) 109055–109067, https://doi.org/10.1016/j.cie.2023.109055.

[84] G.M. George, L.S. Jayashree, Fusion of blockchain-IoT network to improve supply chain traceability using ethermint smart chain: a review, KSII Trans. Internet Inf. Syst. 16 (2022) 3694–3722, https://doi.org/10.3837/tiis.2022.11.013.

[85] M.H. Nasir, J. Arshad, M.M. Khan, M. Fatima, K. Salah, R. Jayaraman, Scalable blockchains — A systematic review, Future Gener. Comput. Syst. 126 (2022) 136–162, https://doi.org/10.1016/j.future.2021.07.035.

[86] E. Hassini, M. Ben-Daya, Z. Bahroun, Modeling the impact of IoT technology on food supply chain operations, Ann. Oper. Res. (2023), https://doi.org/10.1007/s10479-023-05464-6.

[87] M.A. Khan, M.S. Siddiqui, M.K.I. Rahmani, S. Husain, Investigation of big data analytics for sustainable smart city development: an emerging country, IEEe Access. 10 (2022) 16028–16036, https://doi.org/10.1109/ACCESS.2021.3115987.

[88] R. Raj, V. Kumar, P. Verma, Big data analytics in mitigating challenges of sustainable manufacturing supply chain, Oper. Manag. Res. 16 (2023) 1886–1900, https://doi.org/10.1007/s12063-023-00408-6.

[89] A. Gehlot, P.K. Malik, R. Singh, S.V. Akram, T. Alsuwian, Dairy 4.0: Intelligent communication ecosystem for the cattle animal welfare with blockchain and IoT enabled technologies, Appl. Sci. 12 (2022) 7316–7336, https://doi.org/10.3390/app12147316.

[90] M. Gorbunova, P. Masek, M. Komarov, A. Ometov, Distributed ledger technology: State-of-the-Art and current challenges, Comput. Sci. Inf. Syst. 19 (2022) 65–85, https://doi.org/10.2298/CSIS210215037G.

[91] A. Vaghani, K. Sood, S. Yu, Security and QoS issues in blockchain enabled next-generation smart logistic networks: a tutorial, Blockchain: Res. Appl. 3 (2022) 100082–100096, https://doi.org/10.1016/j.bcra.2022.100082.

[92] P. Gargini, F. Balestra, Y. Hayashi, Roadmapping of nanoelectronics for the new electronics industry, Appl. Sci. 12 (2021) 308–328, https://doi.org/10.3390/app12010308.

[93] H.H. Pajooh, S. Demidenko, S. Aslam, M. Harris, Blockchain and 6G-enabled IoT, Inventions 7 (2022) 109–128, https://doi.org/10.3390/inventions7040109.

[94] P. Morella, M.P. Lambán, J. Royo, J.C. Sánchez, Study and analysis of the implementation of 4.0 technologies in the agri-food supply chain: a state of the art, Agronomy 11 (2021), https://doi.org/10.3390/agronomy11122526.

[95] J. Rymarczyk, Technologies, opportunities and challenges of the industrial revolution 4.0: Theoretical considerations, Entrep. Bus. Econ. Rev. 8 (2020) 185–198, https://doi.org/10.15678/EBER.2020.080110.

[96] E.M. Kalogeraki, S. Papastergiou, H. Mouratidis, N. Polemi, A novel risk assessment methodology for SCADA maritime logistics environments, Appl. Sci. (Switzerland) 8 (2018) 1477–1509, https://doi.org/10.3390/app8091477.

[97] I. Konovalenko, A. Ludwig, Event processing in supply chain management – The status quo and research outlook, Comput. Ind. 105 (2019) 229–249, https://doi.org/10.1016/j.compind.2018.12.009.

[98] J. Leveling, L. Weickhmann, C. Nissen, C. Kirsch, Event-driven Architecture for Sensor Data Integration for Logistics Services, in: 2018 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM), IEEE, 2018, pp. 536–540, https://doi.org/10.1109/IEEM.2018.8607460.

[99] J. Turek, B. Ocicka, W. Rogowski, B. Jefmański, The role of Industry 4.0 technologies in driving the financial importance of sustainability risk management, Equilibrium. Quarterly Journal of Economics and Economic Policy 18 (2023) 1009–1044, https://doi.org/10.24136/eq.2023.032.
[100] N. Hasteer, R. Sindhwani, A. Behl, A. Varshney, A. Sharma, Exploring the inhibitors for competitive AI software development through cloud driven transformation, Ann. Oper. Res. (2023), https://doi.org/10.1007/s10479-023-05619-5.
[101] G. Salvini, G.J. Hofstede, C.N. Verdouw, K. Rijswijk, L. Klerkx, Enhancing digital transformation towards virtual supply chains: a simulation game for Dutch floriculture, Prod. Plan. Control. 33 (2022) 1252–1269, https://doi.org/10.1080/09537287.2020.1858361.
[102] A. Hassoun, H.J.P. Marvin, Y. Bouzembrak, F.J. Barba, J.M. Castagnini, N. Pallarés, R. Rabail, R.M. Aadil, S.P. Bangar, R. Bhat, J. Cropotova, S. Maqsood, J. M. Regenstein, Digital transformation in the agri-food industry: recent applications and the role of the COVID-19 pandemic, Front. Sustain. Food Syst. 7 (2023), https://doi.org/10.3389/fsufs.2023.1217813.
[103] P. Empl, G. Pernul, Digital-twin-based security analytics for the internet of things, Information 14 (2023) 95–116, https://doi.org/10.3390/info14020095.
[104] L. Liu, W. Song, Y. Liu, Leveraging digital capabilities toward a circular economy: reinforcing sustainable supply chain management with Industry 4.0 technologies, Comput. Ind. Eng. 178 (2023) 109113–109129, https://doi.org/10.1016/j.cie.2023.109113.
[105] K. Nayal, S. Kumar, R.D. Raut, M.M. Queiroz, P. Priyadarshinee, B.E. Narkhede, Supply chain firm performance in circular economy and digital era to achieve sustainable development goals, Bus. Strategy. Environ. 31 (2022) 1058–1073, https://doi.org/10.1002/bse.2935.
[106] S. Kumar, R.D. Raut, N. Agrawal, N. Cheikhrouhou, M. Sharma, T. Daim, Integrated blockchain and internet of things in the food supply chain: adoption barriers, Technovation 118 (2022) 102589–102874. https://doi.org/10.1016/j.technovation.2022.102589.
[107] E. Taddei, C. Sassanelli, P. Rosa, S. Terzi, Circular supply chains in the era of industry 4.0: A systematic literature review, Comput. Ind. Eng. 170 (2022) 108268–108292, https://doi.org/10.1016/j.cie.2022.108268.
[108] S. Yadav, S. Luthra, D. Garg, Internet of things (IoT) based coordination system in agri-food supply chain: development of an efficient framework using DEMATEL-ISM, Oper. Manag. Res. 15 (2022) 1–27, https://doi.org/10.1007/s12063-020-00164-x.
[109] Y. Maïzi, Y. Bendavid, Hybrid RFID-IoT simulation modeling approach for analyzing scrubs' distribution solutions in operating rooms, Bus. Process Manag. J. 29 (2023) 1734–1761, https://doi.org/10.1108/BPMJ-12-2022-0658.
[110] B. Mohapatra, S. Tripathy, D. Singhal, R. Saha, Significance of digital technology in manufacturing sectors: Examination of key factors during Covid-19, Res. Transp. Econ. 93 (2022) 101134–101148, https://doi.org/10.1016/j.retrec.2021.101134.
[111] A. Navaei, A.A. Taleizadeh, F. Goodarzian, Designing a new sustainable test kit supply chain network utilizing internet of things, Eng. Appl. Artif. Intell. 124 (2023) 106585–106603, https://doi.org/10.1016/j.engappai.2023.106585.
[112] A.K. Yadav, Shweta, D. Kumar, Blockchain technology and vaccine supply chain: Exploration and analysis of the adoption barriers in the Indian context, Int. J. Prod. Econ. 255 (2023) 108716–108736, https://doi.org/10.1016/j.ijpe.2022.108716.
[113] M.H. Naseem, J. Yang, T. Zhang, W. Alam, Utilizing fuzzy AHP in the evaluation of barriers to blockchain implementation in reverse logistics, Sustainability. 15 (2023) 7961, https://doi.org/10.3390/su15107961. –7679.
[114] S. Rajput, S.P. Singh, Industry 4.0 model for integrated circular economy-reverse logistics network, Int. J. Logist. Res. Appl. 25 (2022) 837–877, https://doi.org/10.1080/13675567.2021.1926950.
[115] P. Dutta, R. Chavhan, P. Gowtham, A. Singh, The individual and integrated impact of blockchain and IoT on sustainable supply chains: A systematic review, Supply Chain Forum 24 (2023) 103–126, https://doi.org/10.1080/16258312.2022.2082851.
[116] Z. Mahmood, The Internet of Things in the Industrial Sector, 1st ed., Springer International Publishing, Cham, 2019 https://doi.org/10.1007/978-3-030-24892-5.
[117] H. Yang, S. Kumara, S.T.S. Bukkapatnam, F. Tsung, The internet of things for smart manufacturing: A review, IISe Trans. 51 (2019) 1190–1216, https://doi.org/10.1080/24725854.2018.1555383.
[118] Y. Zuo, Making smart manufacturing smarter–a survey on blockchain technology in industry 4.0, Enterp. Inf. Syst. 15 (2021) 1323–1353, https://doi.org/10.1080/17517575.2020.1856425.
[119] E.J. De Aguiar, B.S. Faiçal, B. Krishnamachari, J. Ueyama, A survey of blockchain-based strategies for healthcare, ACM. Comput. Surv. 53 (2021) 1–27, https://doi.org/10.1145/3376915.
[120] M. Thibaud, H. Chi, W. Zhou, S. Piramuthu, Internet of things (IoT) in high-risk environment, health and safety (EHS) industries: A comprehensive review, Decis. Support. Syst. 108 (2018) 79–95, https://doi.org/10.1016/j.dss.2018.02.005.
[121] A.K. Noon, O. Aziz, I. Zahra, M. Anwar, Implementation of Blockchain in Healthcare: A Systematic Review, in: 4th International Conference on Innovative Computing, ICIC 2021, Institute of Electrical and Electronics Engineers Inc., 2021, https://doi.org/10.1109/ICIC53490.2021.9691510.
[122] W. Viriyasitavat, Z. Bi, D. Hoonsopon, Blockchain technologies for interoperation of business processes in smart supply chains, J. Ind. Inf. Integr. 26 (2022) 100326–100338, https://doi.org/10.1016/j.jii.2022.100326.
[123] J.H. Khor, M. Sidorov, S.A.B. Zulqarnain, Scalable lightweight protocol for interoperable public blockchain-based supply chain ownership management, Sensors 23 (2023) 3433, https://doi.org/10.3390/s23073433. –3250.
[124] A. Rejeb, J.G. Keogh, W. Martindale, D. Dooley, E. Smart, S. Simske, S.F. Wamba, J.G. Breslin, K.Y. Bandara, S. Thakur, K. Liu, B. Crowley, S. Desaraju, A. Ospina, H. Bradau, Charting past, present, and future research in the semantic web and interoperability, Future Internet. 14 (2022) 161–194, https://doi.org/10.3390/fi14060161.
[125] Y. Hu, A. Al-Barakati, P. Rani, Investigating the internet of things (IoT) risks for supply chain management using q-rung orthopair fuzzy-SWARA-ARAS framework, Technol. Econ. Dev. Econ. 30 (2024) 376–401, https://doi.org/10.3846/tede.2022.16583.
[126] K. Yang, T. Duan, J. Feng, A.R. Mishra, Internet of things challenges of sustainable supply chain management in the manufacturing sector using an integrated q-Rung Orthopair Fuzzy-CRITIC-VIKOR method, J. Enterp. Inf. Manag. 35 (2022) 1011–1039, https://doi.org/10.1108/JEIM-06-2021-0261.
[127] Y. He, J. He, N. Wen, The challenges of IoT-based applications in high-risk environments, health and safety industries in the Industry 4.0 era using decision-making approach, J. Innov. Knowl. 8 (2023) 100347–100359, https://doi.org/10.1016/j.jik.2023.100347.
[128] S. Akter, K. Khalil, M. Bayoumi, A Survey on hardware security: current trends and challenges, IEEe Access. 11 (2023) 77543–77565, https://doi.org/10.1109/ACCESS.2023.3288696.
[129] H.A. Abdulghani, N.A. Nijdam, D. Konstantas, Analysis on security and privacy guidelines: RFID-based IoT applications, IEEe Access. 10 (2022) 131528–131554, https://doi.org/10.1109/ACCESS.2022.3227449.
[130] P. Sharma, R. Jindal, M.D. Borah, A review of smart contract-based platforms, applications, and challenges, Cluster. Comput. 26 (2023) 395–421, https://doi.org/10.1007/s10586-021-03491-1.
[131] Y. Zhang, L. Chen, M. Battino, M.A. Farag, J. Xiao, J. Simal-Gandara, H. Gao, W. Jiang, Blockchain: An emerging novel technology to upgrade the current fresh fruit supply chain, Trends. Food Sci. Technol. 124 (2022) 1–12, https://doi.org/10.1016/j.tifs.2022.03.030.
[132] B. Shrimali, H.B. Patel, Blockchain state-of-the-art: architecture, use cases, consensus, challenges and opportunities, J. King Saud Univ. - Comput. Inf. Sci. (2021) 6793–6807, https://doi.org/10.1016/j.jksuci.2021.08.005.
[133] F. Dunke, I. Heckmann, S. Nickel, F. Saldanha-da-Gama, Time traps in supply chains: Is optimal still good enough? Eur J Oper Res 264 (2018) 813–829, https://doi.org/10.1016/j.ejor.2016.07.016.
[134] S. Mithun Ali, M.A. Moktadir, G. Kabir, J. Chakma, M.J.U. Rumi, M.T. Islam, Framework for evaluating risks in food supply chain: Implications in food wastage reduction, J. Clean. Prod. 228 (2019) 786–800, https://doi.org/10.1016/j.jclepro.2019.04.322.
[135] D. Gupta, S. Bhatt, M. Gupta, A.S. Tosun, Future smart connected communities to fight COVID-19 outbreak, IoT (Netherlands) 13 (2021) 100342–100368, https://doi.org/10.1016/j.iot.2020.100342.
[136] S.V. Akram, P.K. Malik, R. Singh, G. Anita, S. Tanwar, Adoption of blockchain technology in various realms: Opportunities and challenges, Security and Privacy 3 (2020), https://doi.org/10.1002/spy2.109.
[137] D. Unal, M. Hammoudeh, M.A. Khan, A. Abuarqoub, G. Epiphaniou, R. Hamila, Integration of federated machine learning and blockchain for the provision of secure big data analytics for internet of things, Comput. Secur. 109 (2021) 102393–102407, https://doi.org/10.1016/j.cose.2021.102393.

[138] L. Xu, S. Mak, A. Brintrup, Will bots take over the supply chain? Revisiting agent-based supply chain automation, Int. J. Prod. Econ. 241 (2021) 108279–108295, https://doi.org/10.1016/j.ijpe.2021.108279.
[139] P.K.R. Maddikunta, Q.V. Pham, P. B, N. Deepa, K. Dev, T.R. Gadekallu, R. Ruby, M. Liyanage, Industry 5.0: A survey on enabling technologies and potential applications, J. Ind. Inf. Integr. 26 (2022) 100257–100278, https://doi.org/10.1016/j.jii.2021.100257.
[140] D. Prajapati, S.K. Jauhar, A. Gunasekaran, S.S. Kamble, S. Pratap, Blockchain and IoT embedded sustainable virtual closed-loop supply chain in E-commerce towards the circular economy, Comput. Ind. Eng. 172 (2022) 108530–108549, https://doi.org/10.1016/j.cie.2022.108530.
[141] A.Z. Abideen, V.P.K. Sundram, J. Pyeman, A.K. Othman, S. Sorooshian, Food supply chain transformation through technology and future research directions—A Systematic Review, Logistics 5 (2021) 83, https://doi.org/10.3390/logistics5040083.
[142] S. Yadav, D. Garg, S. Luthra, Development of IoT based data-driven agriculture supply chain performance measurement framework, J. Enterp. Inf. Manag. 34 (2020) 292–327, https://doi.org/10.1108/JEIM-11-2019-0369.
[143] S. Kumar, R.D. Raut, P. Priyadarshinee, S.K. Mangla, U. Awan, B.E. Narkhede, The impact of IoT on the performance of vaccine supply chain distribution in the COVID-19 context, IEEe Trans. Eng. Manage (2024) 1–11, https://doi.org/10.1109/TEM.2022.3157625.
[144] S. Kumar, R.R. Mallipeddi, Impact of cybersecurity on operations and supply chain management: Emerging trends and future research directions, Prod. Oper. Manage 31 (2022) 4488–4500, https://doi.org/10.1111/poms.13859.
[145] A. Aliahmadi, H. Nozari, Evaluation of security metrics in AIoT and blockchain-based supply chain by neutrosophic decision-making method, Supply Chain Forum 24 (2023) 31–42, https://doi.org/10.1080/16258312.2022.2101898.
[146] J. Gubbi, R. Buyya, S. Marusic, M. Palaniswami, Internet of Things (IoT): a vision, architectural elements, and future directions, Future Gener. Comput. Syst. 29 (2013) 1645–1660, https://doi.org/10.1016/j.future.2013.01.010.
[147] A. Al-Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari, M. Ayyash, Internet of things: A survey on enabling technologies, protocols, and applications, IEEE Commun. Surv. Tut. Journal 17 (2015) 2347–2376, https://doi.org/10.1109/COMST.2015.2444095.
[148] D. Miorandi, S. Sicari, F. De Pellegrini, I. Chlamtac, Internet of things: Vision, applications and research challenges, Ad. Hoc. Netw. 10 (2012) 1497–1516, https://doi.org/10.1016/j.adhoc.2012.02.016.
[149] L.Da Xu, W. He, S. Li, Internet of things in industries: a survey, IEEe Trans. Industr. Inform. 10 (2014) 2233–2243, https://doi.org/10.1109/TII.2014.2300753.
[150] C.N. Verdouw, J. Wolfert, A.J.M. Beulens, A. Rialland, Virtualization of food supply chains with the internet of things, J. Food Eng. 176 (2016) 128–136, https://doi.org/10.1016/j.jfoodeng.2015.11.009.
[151] M.A. Khan, K. Salah, IoT security: Review, blockchain solutions, and open challenges, Future Gener. Comput. Syst. 82 (2018) 395–411, https://doi.org/10.1016/j.future.2017.11.022.
[152] A. Tzounis, N. Katsoulas, T. Bartzanas, C. Kittas, Internet of things in agriculture, recent advances and future challenges, Biosyst. Eng. 164 (2017) 31–48, https://doi.org/10.1016/j.biosystemseng.2017.09.007.
[153] L. Atzori, A. Iera, G. Morabito, The internet of things: A survey, Comput. Netw. 54 (2010) 2787–2805, https://doi.org/10.1016/j.comnet.2010.05.010.
[154] Z. Zheng, S. Xie, H.N. Dai, X. Chen, H. Wang, Blockchain challenges and opportunities: A survey, Int. J. Web. Grid. Services 14 (2018) 352–376, https://doi.org/10.1504/IJWGS.2018.095647.
[155] N. Kshetri, Blockchain's roles in meeting key supply chain management objectives, Int. J. Inf. Manage (2018) 80–89, https://doi.org/10.1016/j.ijinfomgt.2017.12.005.
[156] A. Reyna, C. Martín, J. Chen, E. Soler, M. Díaz, On blockchain and its integration with IoT. Challenges and opportunities, Future Gener. Comput. Syst. 88 (2018) 173–190, https://doi.org/10.1016/j.future.2018.05.046.
[157] S. Saberi, M. Kouhizadeh, J. Sarkis, L. Shen, Blockchain technology and its relationships to sustainable supply chain management, Int. J. Prod. Res. 57 (2019) 2117–2135, https://doi.org/10.1080/00207543.2018.1533261.
[158] A. Kosba, A. Miller, E. Shi, Z. Wen, C. Papamanthou, Hawk: The Blockchain Model of Cryptography and Privacy-Preserving Smart Contracts, in: Proceedings - 2016 IEEE Symposium on Security and Privacy, SP 2016, Institute of Electrical and Electronics Engineers Inc, 2016, pp. 839–858, https://doi.org/10.1109/SP.2016.55.
[159] S.A. Abeyratne, R. Monfared, Blockchain ready manufacturing supply chain using distributed ledger, Int J Res Eng Technol 05 (2016) 1–10, https://doi.org/10.15623/IJRET.2016.0509001.
[160] F. Casino, T.K. Dasaklis, C. Patsakis, A systematic literature review of blockchain-based applications: Current status, classification and open issues, Telemat. Inform. 36 (2019) 55–81, https://doi.org/10.1016/j.tele.2018.11.006.
[161] F. Tian, An agri-food supply chain traceability system for China based on RFID & blockchain technology, in: Proceedings - 2016 International Conference on Service Systems and Service Management, ICSSSM 2016, Institute of Electrical and Electronics Engineers Inc., 2016, pp. 1–6, https://doi.org/10.1109/ICSSSM.2016.7538424.
[162] J.F. Galvez, J.C. Mejuto, J. Simal-Gandara, Future challenges on the use of blockchain for food traceability analysis, TrAC - Trends in Analytical Chemistry 107 (2018) 222–232, https://doi.org/10.1016/j.trac.2018.08.011.
[163] K. Christidis, M. Devetsikiotis, Blockchains and smart contracts for the internet of things, IEEe Access. 4 (2016) 2292–2303, https://doi.org/10.1109/ACCESS.2016.2566339.
[164] G. Zhao, S. Liu, C. Lopez, H. Lu, S. Elgueta, H. Chen, B.M. Boshkoska, Blockchain technology in agri-food value chain management: A synthesis of applications, challenges and future research directions, Comput. Ind. 109 (2019) 83–99, https://doi.org/10.1016/j.compind.2019.04.002.
[165] A. Kamilaris, A. Fonts, F.X. Prenafeta-Boldú, The rise of blockchain technology in agriculture and food supply chains, Trends. Food Sci. Technol. 91 (2019) 640–652, https://doi.org/10.1016/j.tifs.2019.07.034.
[166] S.S. Kamble, A. Gunasekaran, R. Sharma, Modeling the blockchain enabled traceability in agriculture supply chain, Int. J. Inf. Manage 52 (2020) 101967–101983, https://doi.org/10.1016/j.ijinfomgt.2019.05.023.
[167] K. Salah, N. Nizamuddin, R. Jayaraman, M. Omar, Blockchain-based soybean traceability in agricultural supply chain, IEEe Access. 7 (2019) 73295–73305, https://doi.org/10.1109/ACCESS.2019.2918000.